From Snapshot to Movie

Module 5.1: Panel Survey Design Fundamentals

Day 5: Synthesis, Longitudinal Surveys, and Conclusion

SADC Regional Training Workshop 2026

The Minister's Challenge

Zambara Case

The Household Living Conditions Survey was a success. But the Minister poses a harder question...

*"Your survey tells us where we are **today**.*
*But how do we know if our policies are **working**?"*

— Minister of Finance

▶ Key Insight

A cross-sectional survey is a **photograph**. To measure **real change**, we need a **movie**.

Learning Objectives

By the end of this module, you will be able to:

1. **Distinguish** between cross-sectional and panel survey designs
2. **Explain** why "gross change" matters more than "net change" for policy
3. **Design** sample rotation patterns for quarterly surveys
4. **Calculate** and interpret attrition rates by subgroup
5. **Implement** panel data linking using unique identifiers

Reference: Lynn, P. (2009). *Methodology of Longitudinal Surveys*. Wiley.

The Fundamental Question

What Cross-Sectional Tells Us

Poverty rate in 2025: 25%
Poverty rate in 2026: 24%

The Gap →

What We Don't Know

Are they the *same* poor?
Or different households?

Net Change (Cross-Sectional)

$$25\% \rightarrow 24\% = -1\%$$

"Poverty fell by 1 percentage point"

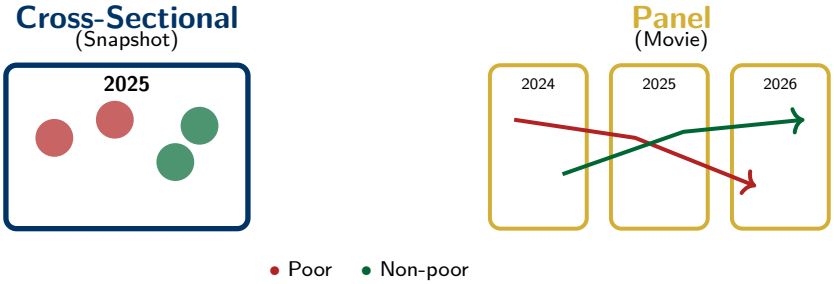
Gross Change (Panel)

8% escaped poverty
7% fell into poverty

Net: -1%**, Gross:** 15%

Source: Baulch & Hoddinott (2000). "Economic mobility and poverty dynamics." *Journal of Development Studies*.

Snapshot vs. Movie: A Visual Concept



▶ Key Insight

Panels track the **same units** over time, revealing **trajectories** not just states.

Why Net Change Can Deceive

Two Countries, Same Net Change, Very Different Realities

Country A: "Stable"

2025: 30% poor
2026: 30% poor

5% escaped
5% fell in
Net: 0%, Gross: 10%

Country B: "Churning"

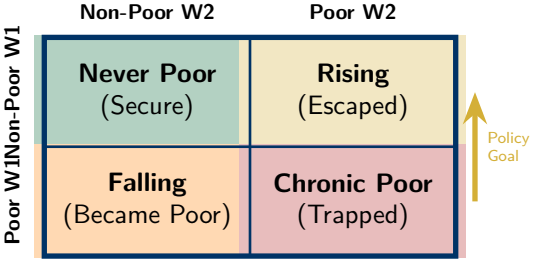
2025: 30% poor
2026: 30% poor

20% escaped
20% fell in
Net: 0%, Gross: 40%

Policy Implication: Country B has massive vulnerability despite "stable" poverty rate!

Source: Dercon & Shapiro (2007). "Moving on, staying behind, getting lost." *Global Poverty Research Group*.

Poverty Dynamics: What Panels Reveal



▶ Key Insight

Only panels can distinguish **chronic poverty** (trapped) from **transient poverty** (temporary shock). Different problems require different policies!

What is a Panel Survey?

Panel Survey Definition

A **panel survey** (or longitudinal survey) collects data from the **same sample units** at multiple points in time, enabling measurement of **individual-level change**.

Key Characteristics:

- Same units tracked over time
- Unique identifiers link records
- Multiple "waves" of data collection
- Can measure gross flows
- Enables causal inference

Famous Examples:

- PSID (USA, since 1968)
- BHPS/UKHLS (UK)
- NIDS (South Africa)
- EU-SILC (Europe)
- QLFS rotation panels

Reference: Hsiao, C. (2014). *Analysis of Panel Data*. 3rd ed. Cambridge University Press.

Cross-Sectional vs. Panel: A Comparison

Feature	Cross-Sectional	Panel
Sample units	Different each wave	Same over time
Change	Net change only	Net + Gross
Causal inference	Limited	Stronger
Attrition risk	None	Significant
Cost per wave	Lower	Higher

△ Caution

Panels are powerful but demanding. Attrition and seam bias require careful management.

The Statistical Power of Panels

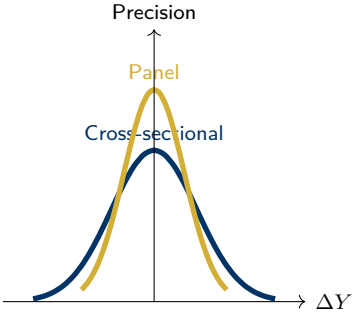
Why panels are more efficient:

For measuring **change** $(\bar{Y}_2 - \bar{Y}_1)$:

$$\text{Var}(\bar{Y}_2 - \bar{Y}_1) = \text{Var}(\bar{Y}_2) + \text{Var}(\bar{Y}_1) - 2\text{Cov}(\bar{Y}_1, \bar{Y}_2)$$

With the **same** sample:

- Covariance is **positive**
- Variance of difference is **reduced**
- Time-invariant factors "difference out"



Panel estimates of change are **more precise**

Reference: Kish, L. (1965). *Survey Sampling*. Chapter 12. Wiley.

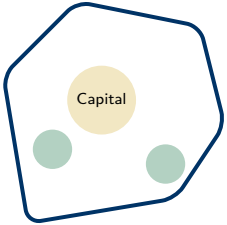
Zambara Case: The QLFS Pilot

Zambara Case

Lindiwe proposes the Zambara QLFS—a rotating panel to track employment transitions.

ZQLFS Design Parameters:

- Quarterly data collection
- 20,940 households in Wave 1
- 2-2-2 rotation pattern
- Province-level estimates



Zambara

Key Question: What happens when households drop out between waves?

What is a Rotation Pattern?

Rotation Pattern

A systematic scheme for replacing portions of the sample over time, balancing the need for **continuity** (tracking same units) with **freshness** (new sample units).

Why rotate?

✓ **Benefits of Continuity:**

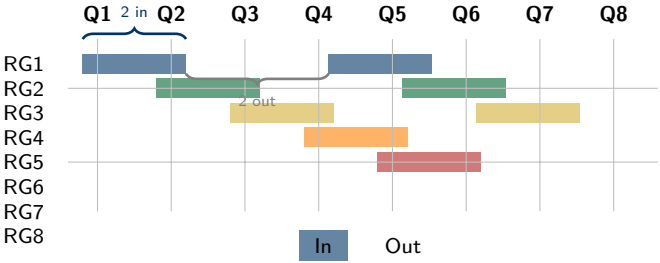
- Measure individual change
- Higher precision for trends
- Control for unobserved heterogeneity

✓ **Benefits of Freshness:**

- Reduce respondent fatigue
- Limit conditioning effects
- Maintain representativeness

Reference: Kalton, G. & Citro, C. (1995). "Panel surveys." *Journal of Official Statistics*.

The 2-2-2 Rotation Pattern



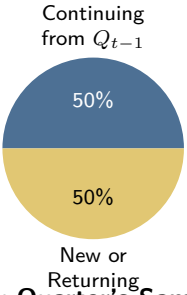
▶ Key Insight

Each household: **2 quarters in**, rests **2 quarters**, returns **2 more**, then exits.

Sample Overlap Under 2-2-2 Pattern

Quarter-to-Quarter Overlap:

Comparison	Overlap
Q_t vs Q_{t+1}	50%
Q_t vs Q_{t+2}	0%
Q_t vs Q_{t+3}	0%
Q_t vs Q_{t+4}	50%



Any Quarter's Sample

Implication: Year-over-year comparisons (Q_t vs Q_{t+4}) benefit from 50% overlap—ideal for annual employment change estimates.

Reference: Hansen, M.H. et al. (1955). "Sample survey methods and theory." Wiley.

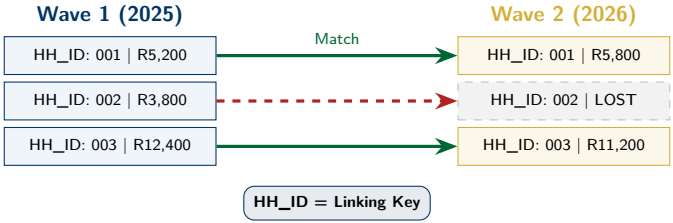
Alternative Rotation Patterns

Pattern	Used By	Overlap	Best For
2-2-2	EU-LFS, African LFS	50% Q-Q, 50% YoY	Quarterly labour
4-8-4	US CPS	75% M-M, 50% YoY	Monthly employment
Pure Panel	PSID, NIDS	100%	Long-term dynamics
No Rotation	Cross-sections	0%	Point estimates

Zambara Case

Lindiwe chooses 2-2-2 for ZQLFS, matching Statistics South Africa's QLFS approach.

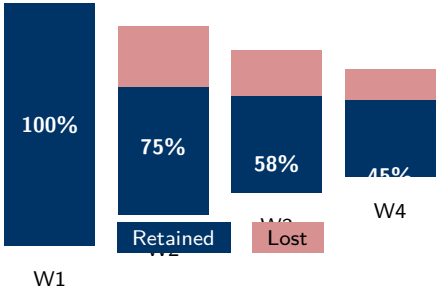
The "Time-Travel" Key: Unique Identifiers



▶ Key Insight

The unique identifier (e.g., `UqNr`) is the "time-travel" key connecting the **same household** across waves. Without it, panel analysis is impossible!

Attrition: The Panel's Greatest Enemy



△ Caution

Cumulative attrition compounds over waves. A 25% attrition rate per wave leaves only 42% of the original sample after 4 waves!

Types of Attrition

Unit Non-Response

- **Non-contact:** Moved, not home
- **Refusal:** Declined to participate
- **Inability:** Illness, death
- **Dissolution:** Household no longer exists

Location-Based

- **Migration:** Moved out of area
- **Inaccessible:** Conflict, disaster
- **Address error:** Wrong location recorded
- **Demolished:** Structure no longer exists

Critical Question: Is attrition random or systematic?
Systematic attrition creates **selection bias** in panel estimates.

Zambara ZQLFS: Wave 1 Results

Zambara Case

The first wave of ZQLFS was fielded in Q1 2026. Here is the achieved sample.

Province	Wave 1 (N)	%
Zambara Capital	5,526	26.4%
Eastern Highlands	3,257	15.6%
Coastal Plains	2,706	12.9%
Northern Bushveld	2,385	11.4%
Southern Cape	1,931	9.2%
Other provinces	5,135	24.5%
TOTAL	**20,940**	**100%**

Note: Other provinces = Eastern Forests, Mining Belt, Central Plateau, Western Drylands

Wave 2: The Attrition Shock

Zambara Case

Wave 2 fieldwork reveals: **5,225 households** (25.0%) could not be re-interviewed!

Overall Results:

Wave 1 Sample	20,940
Wave 2 Sample	15,715
Attrited	5,225
Attrition Rate	**25.0%**

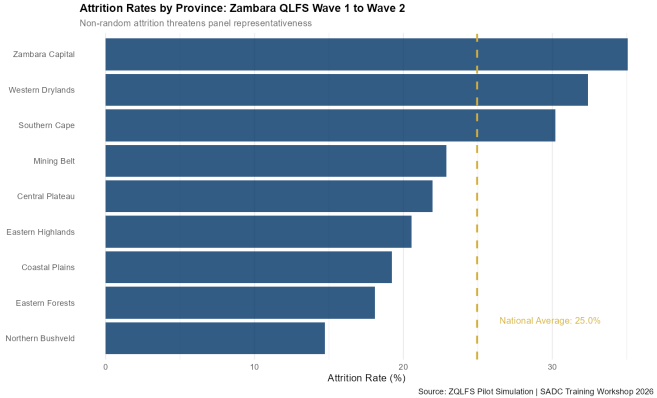
One in four households lost!

Attrition by Province: A Clear Pattern Emerges

Province	Wave 1	Wave 2	Attrited	Rate
Zambara Capital	5,526	3,588	1,938	**35.1%**
Western Drylands	855	578	277	**32.4%**
Southern Cape	1,931	1,348	583	**30.2%**
Mining Belt	1,333	1,028	305	22.9%
Central Plateau	1,276	996	280	21.9%
Eastern Highlands	3,257	2,588	669	20.5%
Coastal Plains	2,706	2,186	520	19.2%
Eastern Forests	1,671	1,369	302	**18.1%**
Northern Bushveld	2,385	2,034	351	**14.7%**
TOTAL	**20,940**	**15,715**	**5,225**	**25.0%**

Source: ZQLFS Pilot Simulation (Lab 5.1) | Verified output from R script

Visualizing Non-Random Attrition



Source: ZQLFS Pilot Simulation (Lab 5.1)

Why Does Attrition Vary by Province?

High Attrition Regions:

Zambara Capital (35.1%)

- High residential mobility
- Young, mobile population

Western Drylands (32.4%)

- Sparse settlements
- Seasonal migration

Low Attrition Regions:

Northern Bushveld (14.7%)

- Stable rural communities
- Strong traditional ties

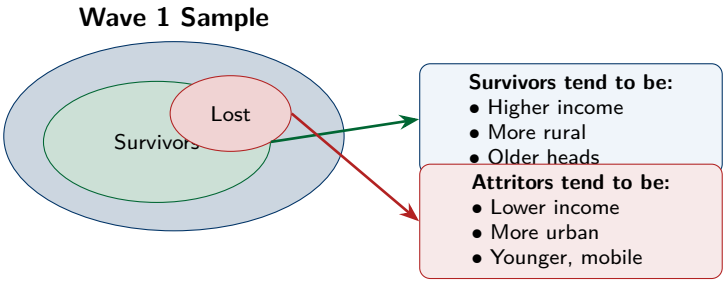
Eastern Forests (18.1%)

- Agricultural settlements
- Lower refusal rates

△ Caution

Non-random attrition means Wave 2 **underrepresents** urban and mobile populations!

Selection Bias: Survivors $\neq$ Population



▶ Key Insight

Analyzing only survivors without adjustment **overestimates** income and **underestimates** poverty!

Panel Data Linking: The Merge Operation

Wave 1

hh_id	inc_w1
001	5,200
002	3,800
003	12,400

MERGE →

Wave 2

hh_id	inc_w2
001	5,800
003	11,200

Linked Panel (N=15,715)

hh_id	inc_w1	inc_w2	chg
001	5,200	5,800	+600
003	12,400	11,200	-1,200

R Code: Panel Data Linking

Panel Merge in R

```
# Link panels using unique household identifier
panel_linked <- wave1 %>%
  inner_join(wave2, by = "hh_id")

# Calculate change
panel_linked <- panel_linked %>%
  mutate(income_change = income_w2 - income_w1,
         pct_change = (income_change/income_w1)*100)
# Result: 15,715 linked households
```

The `inner_join` keeps only households present in **both** waves (balanced panel).

STATA Code: Panel Data Linking

Panel Merge in STATA

```
* Load Wave 1 and merge with Wave 2
use "zambara_wave1.dta", clear
keep hh_id income_w1 province

merge 1:1 hh_id using "zambara_wave2.dta", ///
      keepusing(income_w2) keep(match) nogen

* Calculate change
gen income_change = income_w2 - income_w1
gen pct_change = (income_change/income_w1)*100
* Result: 15,715 linked households
```

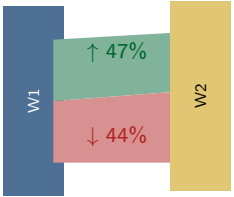
The `keep(match)` option retains only observations that merge successfully.

Income Dynamics in the Linked Panel

Change Summary:

Metric	Value
Mean Income W1	R 6,842
Mean Income W2	R 7,015
Mean Change	R +173
% Increased	47.3%
% Decreased	44.2%

Net: +2.5% | Gross: 91.5%



Massive churning beneath a stable average!

▶ Key Insight

Net change (+2.5%) masks enormous **gross flows**: nearly all households moved.

Module 5.1: Key Takeaways

1. **Panels reveal what cross-sections hide:** Gross change, transitions, individual trajectories—not just aggregate snapshots.
2. **Rotation patterns balance competing goals:** The 2-2-2 pattern provides 50% overlap while limiting respondent burden.
3. **Attrition is the panel's enemy:** 25% attrition in ZQLFS Wave 2, with systematic patterns by region.
4. **Non-random attrition creates bias:** Survivors differ from attritors—weighting adjustments are essential.
5. **The unique identifier is sacred:** Without `hh_id`, panel linking is impossible.

Coming Up: Lab 5.2 — Attrition Weight Adjustment

Zambara Case

Lindiwe now faces a critical question: How can we adjust for the non-random loss of households to maintain representativeness?

In Lab 5.2, you will:

- Build a **response propensity model** predicting attrition
- Calculate **inverse probability weights** for survivors
- Compare weighted vs. unweighted estimates
- Assess **bias reduction** from attrition adjustment

Module 5.1 Panel Design → **Lab 5.2 Attrition Weights** → Module 5.3 Weight Types

Rotation Pattern Design

Slides 31–60

Balancing Continuity with Freshness

Day 5 | Module 5.1 Part 2

The Rotation Design Challenge

Zambara Case

Lindiwe must design a rotation pattern for ZQLFS. She faces competing objectives...

Continuity (Overlap)

- Track same units over time
- Measure gross flows
- Reduce variance of change
- Enable longitudinal analysis

Freshness (Turnover)

- Reduce respondent burden
- Limit conditioning effects
- Maintain representativeness
- Refresh aging sample

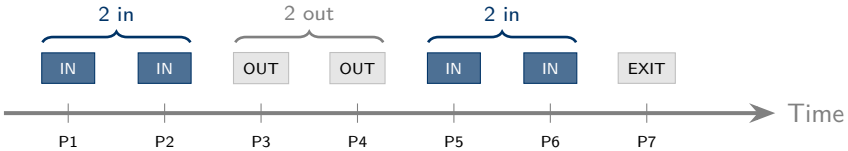
▶ Key Insight

The rotation pattern determines how these objectives are balanced—there is no single "best" design.

Anatomy of a Rotation Pattern

Rotation Pattern Notation

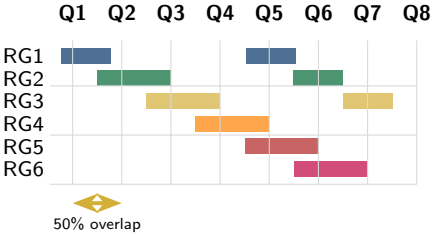
A-B-C means: **A** periods in sample, **B** periods out, **C** periods back in



Example: 2-2-2 Pattern

Total time in panel: $A + C = 4$ periods. Total panel duration: $A + B + C = 6$ periods.

The 2-2-2 Pattern: Full Visualization



In any quarter, 4 rotation groups are active (2 in first stint, 2 in second stint).
Result: 50% quarter-to-quarter overlap, 50% year-over-year overlap.

The 4-8-4 Pattern: US Current Population Survey

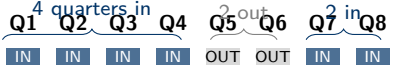
Advantages:

- High monthly overlap (75%)
- Good year-over-year comparison
- Manageable respondent burden

Used by:

- US Current Population Survey
- Canadian Labour Force Survey
- Some Latin American LFS

The 4-2-2 Pattern: An Alternative



Pattern	Q-to-Q Overlap	YoY Overlap	Total Interviews
2-2-2	50%	50%	4
4-2-2	75% (first year)	50%	6
4-8-4	75%	50%	8

▶ Key Insight

4-2-2 provides higher overlap in the first year but more interviews overall.

Calculating Sample Overlap

Overlap Formula

For a rotation pattern with R rotation groups and k groups replaced each period:

$$\text{Period-to-Period Overlap} = \frac{R-k}{R} \times 100\%$$

Example: 2-2-2 Pattern

- $R = 4$ active rotation groups per quarter
- $k = 2$ groups rotate out, 2 new groups rotate in
- Overlap $= \frac{4-2}{4} = 50\%$

Example: 4-8-4 Pattern

- $R = 8$ active rotation groups per month
- $k = 2$ groups rotate out each month
- Overlap $= \frac{8-2}{8} = 75\%$

Why Overlap Matters: Variance Reduction

For estimating **change** $\Delta = \bar{Y}_t - \bar{Y}_{t-1}$:

Variance of Change Estimate

$$\text{Var}(\hat{\Delta}) = \text{Var}(\bar{Y}_t) + \text{Var}(\bar{Y}_{t-1}) - 2\rho \cdot \text{SD}(\bar{Y}_t) \cdot \text{SD}(\bar{Y}_{t-1})$$

Where:

- ρ = correlation between estimates
- Higher overlap $\Rightarrow$ higher ρ
- Higher $\rho \Rightarrow$ lower variance

With 50% overlap and $\rho = 0.7$:
Variance reduced by $\approx 30\%$

Overlap	Var. Red.
25%	15%
50%	30%
75%	45%

(Assuming $\rho \approx 0.7$)

Reference: Kish, L. (1965). *Survey Sampling*. Chapter 12.

The AK Composite Estimator

Zambara Case

Lindiwe learns that overlapping samples enable powerful composite estimators...

AK Composite Estimator

$$\hat{Y}_t^{AK} = K \cdot \hat{Y}_t + (1 - K) \cdot (\hat{Y}_{t-1}^{AK} + \hat{\Delta}_t)$$

Components:

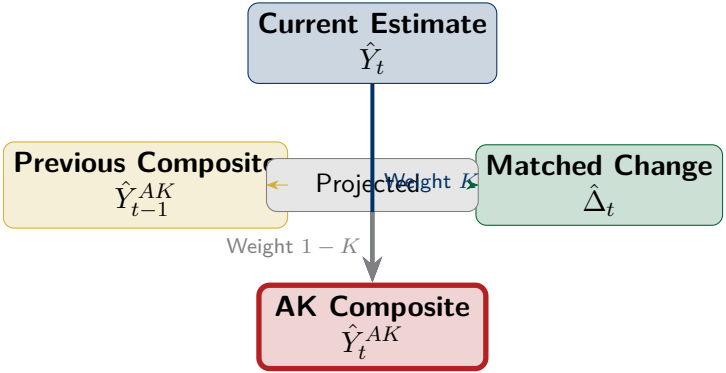
- $\hat{Y}_t$ = current period estimate
- $\hat{Y}_{t-1}^{AK}$ = previous composite
- $\hat{\Delta}_t$ = matched sample change
- K = weighting factor (typically 0.4–0.6)

Benefits:

- Reduces sampling variance
- Smoother time series
- Better month-to-month
- Used by US BLS

Reference: Gurney & Daly (1965). "A Multivariate Approach to Estimation." *ASA Proceedings*.

AK Estimator: Visual Intuition

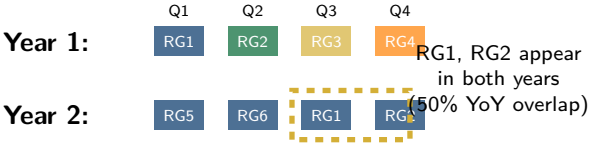


The composite blends current data with projected estimate from matched panel.

Year-over-Year Overlap: Seasonal Comparisons

Zambara Case

The Minister wants to compare Q1 2026 to Q1 2025. Can the rotation design help?



▶ Key Insight

The 2-2-2 pattern provides 50% YoY overlap, enabling matched comparisons of seasonal data for unemployment trends.

Rotation Design Trade-offs

Criterion	2-2-2	4-8-4	4-2-2	Panel
Q-to-Q overlap	50%	75%	75%	100%
YoY overlap	50%	50%	50%	100%
Respondent burden	Low	Medium	Medium	High
Attrition impact	Low	Medium	Medium	High
Gross flows	Limited	Good	Good	Full
Sample freshness	High	Medium	Medium	Low

△ Caution

There is no universally optimal pattern—choice depends on survey objectives and constraints.

Conditioning Effects: The Hidden Cost of Continuity

Conditioning Effect

Systematic change in respondent behavior due to repeated interviewing—answers in later waves may differ from first-wave answers even if true status unchanged.

Common manifestations:

- Shortened responses over time
- "Satisficing" behavior
- Learning survey expectations
- Reporting fatigue

Mitigation strategies:

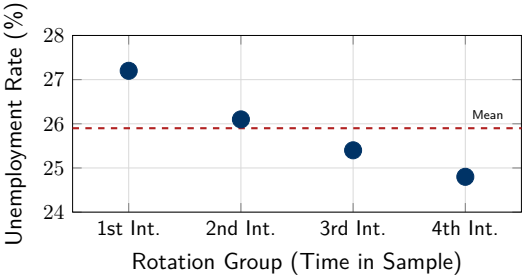
- Limit time in sample
- Rest periods (the "out" phase)
- Incentives for later waves
- Monitor rotation group bias

Reference: Silberstein & Jacobs (1989). "Symptoms of Repeated Interview Effects." *Survey Methodology*.

Rotation Group Bias: A Diagnostic Tool

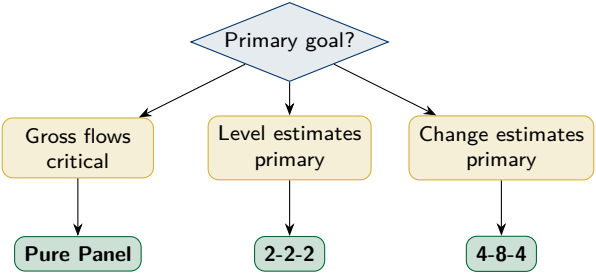
Zambara Case

Lindiwe checks if estimates differ systematically by rotation group...



Declining unemployment with time in sample suggests conditioning—respondents learn to report "employed" more readily.

Lindiwe's Decision Framework

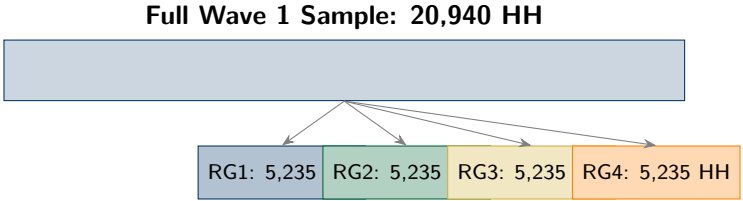


▶ Key Insight

For ZQLFS, Lindiwe chooses **2-2-2**—balancing quarterly level estimates with manageable respondent burden.

Implementing Rotation: The Mechanics

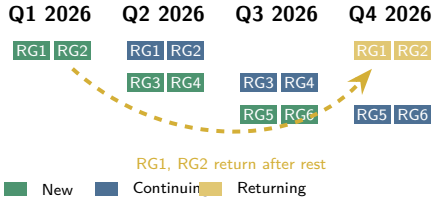
Step 1: Divide sample into rotation groups



Key requirements:

- Each RG must be **independently representative**
- Stratification applied **within** each RG
- Random assignment to rotation groups at PSU level

Sample Flow: First Year of ZQLFS



After ramp-up in Q1, each quarter has 4 active rotation groups.

Handling Ramp-Up and Ramp-Down

Ramp-Up (Survey Start):

- Q1: Only 2 RGs (half sample)
- Q2: Full 4 RGs achieved
- Alternative: Start all 4 RGs at once

Implication: Q1 estimates have higher variance or require adjustment.

Ramp-Down (Survey End):

- Stop adding new RGs
- Let existing RGs complete
- Sample shrinks gradually

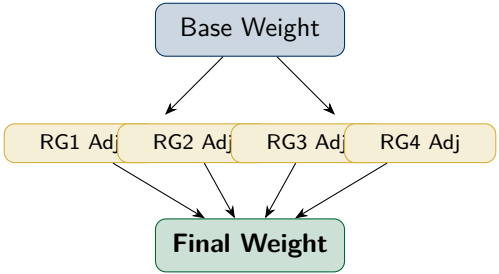
Implication: Final quarters have reduced precision.

△ Caution

Many agencies ignore ramp-up effects and publish Q1 with caveats. Plan ramp-up timing carefully!

Weights Under Rotation

Each rotation group needs its own weight adjustment:



Weights must account for: (1) differential non-response by RG, (2) attrition in returning RGs, (3) calibration to current population totals.

Two Types of Weights

Cross-Sectional Weights

- Used for: Point-in-time estimates
- Goal: Represent current population
- Calibrated: Each wave separately
- Example: "25% unemployed in Q2"

All responding units in wave t receive weights summing to population.

Longitudinal Weights

- Used for: Change/transition analysis
- Goal: Represent Wave 1 population
- Adjusted: For attrition each wave
- Example: "12% moved out of poverty"

Only units in all waves; weights adjusted for selective attrition.

▶ Key Insight

Cross-sectional and longitudinal weights serve different purposes—the same survey often provides both!

Rotation Patterns Across SADC

Country	Survey	Pattern	Notes
South Africa	QLFS	2-2-2	Since 2008
Botswana	QLFS	2-2-2	Quarterly since 2021
Namibia	LFS	Annual rotation	50% overlap
Mauritius	CMPHS	2-2-2	Monthly collection
Zimbabwe	LFS	No rotation	Cross-sectional
Tanzania	LFS	Partial panel	Annual

Zambara Case

Following regional best practice, Lindiwe confirms the 2-2-2 pattern for ZQLFS, aligning with Stats SA methodology.

Common Implementation Pitfalls

1. **Unbalanced rotation groups**
   - ✗ RGs differ systematically in size or composition
   - ✓ Random assignment at PSU level with stratification
2. **Ignoring ramp-up effects**
   - ✗ Publishing first quarter without caveats
   - ✓ Document reduced precision or delay publication
3. **Losing track of rotation status**
   - ✗ Unclear which households are in which stint
   - ✓ Maintain rotation group variable in data
4. **Applying wrong weights**
   - ✗ Using cross-sectional weights for transitions
   - ✓ Create purpose-specific weight variables

Data Structure: Key Variables

Essential Rotation Variables

hh_id	wave	rot_grp	stint	active	
001	2026Q1	1	1	1	
001	2026Q2	1	1	1	
001	2026Q3	1	.	0	(out)
001	2026Q4	1	.	0	(out)
001	2027Q1	1	2	1	(return)
001	2027Q2	1	2	1	
001	2027Q3	1	.	0	(exit)

`rot_grp` = rotation group assignment (fixed).
`stint` = 1st or 2nd period in sample.
`active` = currently being interviewed.

Quality Assurance Checklist

Monthly/Quarterly checks:

- ☐ Response rates by RG
- ☐ Key estimates by RG
- ☐ Attrition patterns
- ☐ Conditioning effects test
- ☐ Sample composition by RG
- ☐ Weight distributions
- ☐ Variance by RG
- ☐ Overlap achieved

Red flags:

- Unemployment differs by >2pp across rotation groups
- Response rates differ by >5pp across rotation groups
- Actual overlap deviates from design by >10pp

Lab 5.1 Preview: Rotation Pattern Simulation

Zambara Case

In Lab 5.1, you will simulate 8 quarters of ZQLFS data under different rotation patterns.

Lab objectives:

1. Create rotation group assignments for a sample
2. Simulate 2-2-2 rotation over 8 quarters
3. Calculate achieved overlap percentages
4. Compare variance of change estimates
5. Implement basic composite estimator

Tools: R (`dplyr`, `survey`) and STATA
Data: ZQLFS Wave 1 (20,940 households)

R Code: Assigning Rotation Groups

Rotation Group Assignment

```
# Assign rotation groups randomly within strata
library(dplyr)

zqlfs_w1 <- zqlfs_w1 %>%
  group_by(stratum) %>%
  mutate(
    rot_grp = sample(rep(1:4, length.out = n()))
  ) %>%
  ungroup()

# Check balance
table(zqlfs_w1$rot_grp, zqlfs_w1$province)
```

Each rotation group should be representative: check distributions match.

STATA Code: Calculating Overlap

Overlap Calculation

```
* Merge consecutive quarters
use "zqlfs_q1.dta", clear
merge 1:1 hh_id using "zqlfs_q2.dta", keep(match)

* Count matched households
count if _merge == 3
local matched = r(N)

* Calculate overlap percentage
quietly count using "zqlfs_q1.dta"
display "Q-to-Q overlap: " (`matched'/`q1_n')*100 "%"
```

Expected: ∼50% for 2-2-2 pattern.

Module 5.1 Summary: Key Takeaways

1. **Rotation patterns** balance continuity (overlap) with freshness (turnover)
2. **Common patterns:** 2-2-2 (EU-LFS), 4-8-4 (US CPS), pure panel (PSID)
3. **Higher overlap** $\Rightarrow$ lower variance for change, but higher burden
4. **AK composite estimator** exploits overlap for variance reduction
5. **Conditioning effects** require monitoring (rotation group bias)
6. **Two weight types:** cross-sectional vs. longitudinal

▶ Key Insight

For ZQLFS, Lindiwe selected **2-2-2**: 50% overlap, manageable burden, Stats SA compatible.

References and Further Reading

Core methodology:

- Kish, L. (1965). *Survey Sampling*. Wiley. [Chapter 12]
- Kalton, G. & Citro, C. (1995). "Panel Surveys: Adding the Fourth Dimension."

Composite estimation:

- Gurney, M. & Daly, J. (1965). "Multivariate Approach to Estimation."
- Lent, J. et al. (1999). "Composite Estimation for the CPS."

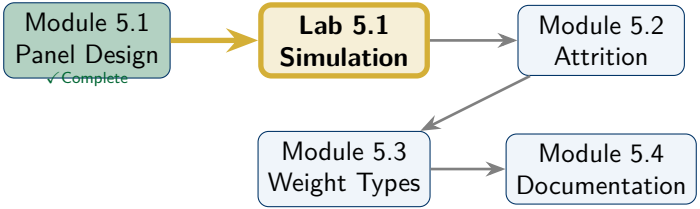
Conditioning effects:

- Silberstein, A. & Jacobs, C. (1989). "Symptoms of Repeated Interview Effects."

SADC practice:

- Statistics South Africa. *QLFS Methodology*. Stats SA.

Coming Up Next



Zambara Case

With the rotation pattern designed, Lindiwe now turns to the biggest threat: **attrition**. How can she adjust for the 25% of households lost between waves?

The Dangers of Merging

Split Households, Movers, and Data Quality

Day 5, Module 5.1 (continued)

Lab 5.1b: Panel Data Linking Complications

The Data Manager's Warning

Zambara ZQLFS Data Office

"We merged Wave 1 and Wave 2 by household ID and got 14,485 matched records. But something's wrong—some IDs appear twice, some don't match at all, and we have more households than we started with in some districts!"

The Hidden Complexities of Panel Linking:

1. Households **split** — one becomes two or more
2. Households **move** — same people, different address
3. IDs get **corrupted** — transcription and matching errors
4. Records get **duplicated** — data entry mistakes

▶ Key Insight

A "successful merge" is not the same as a "correct merge."

Learning Objectives: Linking Data

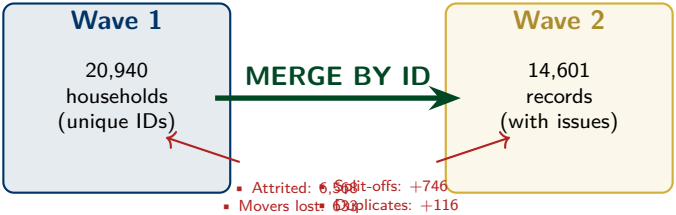
By the end of this section, you will be able to:

1. **Diagnose** common panel merge failures and their causes
2. **Implement** split household tracking protocols
3. **Design** mover tracking strategies
4. **Calculate** true merge success rates (not naive rates)
5. **Compare** different "following rules" and their implications

Reference

Himelein, K. (2014). "Weight Calculations for Panel Surveys with Subsampling and Split-off Tracking." *Statistics and Public Policy*, 1(1), 40–45.

The Merge Operation: What Could Go Wrong?



Four types of merge problems:

- Missing matches — households that disappeared
- Extra records — splits and duplicates
- False non-matches — ID errors breaking links
- False matches — wrong households linked

Taxonomy of Household Fates

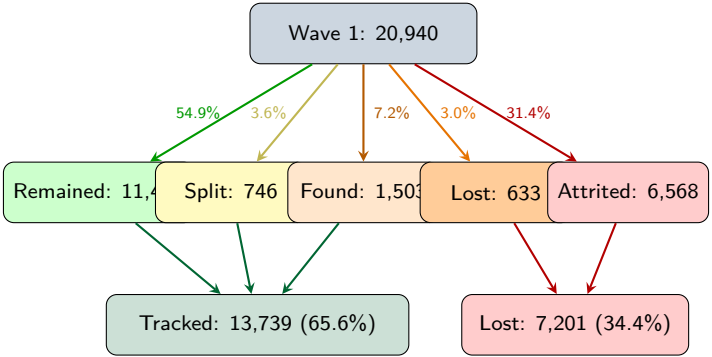
What happens to households between waves?

Fate	Description	ZQLFS Results
Remained	Same HH, same location, interviewed	11,490 (54.9%)
Split	HH divided into 2+ new HHs	746 (3.6%)
Moved (found)	HH relocated, successfully traced	1,503 (7.2%)
Moved (lost)	HH relocated, not found	633 (3.0%)
Attrited	Refused, deceased, other loss	6,568 (31.4%)
Total		20,940 (100%)

▶ Key Insight

Only 54.9% of households have "clean links" — the rest require special handling.

Household Fates: Visual Flow



What is a Split Household?

Definition

A **split household** occurs when one or more members of an original sample household move out and form a new, separate household.

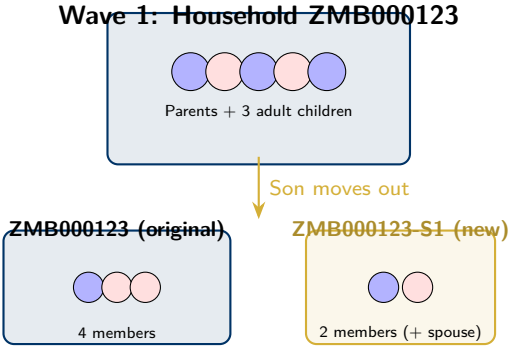
Common causes of household splits:

- Adult children leaving home (marriage, employment)
- Divorce or separation
- Elderly parents moving to different care arrangements
- Economic migration of working-age members

In Zambara ZQLFS:

- 746 original households split (3.6% of Wave 1)
- Each split created 1 new household $\rightarrow$ 746 additional records
- Total split-related records: 1,492 (746 original + 746 new)

The Split Household Problem



△ Caution

If we only track the original ID, we lose the split-off household entirely!

Split Household: ID Convention

Standard ID Naming for Splits

Record Type	ID Format	Example
Original household	Base ID unchanged	ZMB000123
First split-off	Base ID + "-S1"	ZMB000123-S1
Second split-off	Base ID + "-S2"	ZMB000123-S2

R Code for Creating Split IDs:

```
# For split-off households
split_offs <- split_hh %>%
  mutate(
    hh_id_w2 = paste0(hh_id_w1, "-S1"),
    link_status = "Split - New Household",
    parent_hh_id = hh_id_w1  # Track origin
  )
```

Movers: The Tracking Challenge

Definition

A **mover** is a household that relocates to a different dwelling unit between survey waves, while maintaining its composition.

Mover statistics from ZQLFS:

- 2,136 households moved between waves (10.2%)
- 1,503 were successfully traced (70.4% trace rate)
- 633 were lost (29.6% of movers)

Why movers matter:

- Mobile populations often differ systematically from stayers
- Younger, urban, rental households more likely to move
- Losing movers creates **selection bias**

Mover Tracking Protocol

Best Practices for Mover Tracking

1. **Collect tracking information at baseline:**
   - Mobile phone numbers (multiple)
   - Alternative contact persons
   - Employer information
2. **Field protocol for non-contacts:**
   - Query neighbors at original address
   - Contact alternative persons
   - Check with local authorities
3. **Define search radius:** How far to follow?
4. **Budget for tracking:** Mover interviews cost 2–3× more

Reference: World Bank LSMS guidelines recommend minimum 3 tracking contacts per mover household.

The Cost of Not Tracking Movers

WITH Mover Tracking:

- Tracked: 13,739 (65.6%)
- Includes 1,503 found movers
- Maintains representativeness
- Higher cost, better quality

WITHOUT Mover Tracking:

- Tracked: 12,236 (58.4%)
- Loses ALL 2,136 movers
- Systematic bias
- Lower cost, biased estimates

△ Caution

Not tracking movers loses 7.2 percentage points of the sample and systematically excludes mobile populations (young, urban, renters).

Data Quality Issue #1: ID Errors

ID Matching Errors

Sources of ID errors:

- Transcription mistakes (handwriting to digital)
- Digit transposition ("123456" → "123465")
- Leading zero problems ("000123" → "123")
- Different ID systems between waves

ZQLFS Impact:

- 290 records with ID errors (2.0% of Wave 2)
- These create **false non-matches** in the merge
- Records exist in both waves but don't link

Diagnosing ID Errors

R Code for ID Error Detection:

```
# Check for near-matches using string distance
library(stringdist)

# Find Wave 2 IDs not in Wave 1
unmatched_w2 <- wave2_ids[!wave2_ids %in% wave1_ids]

# For each unmatched, find closest match in Wave 1
for (id in unmatched_w2) {
  distances <- stringdist(id, wave1_ids, method = "lv")
  closest <- wave1_ids[which.min(distances)]
  min_dist <- min(distances)

  if (min_dist <= 2) {  # Likely transcription error
    cat(sprintf("Possible match: %s -> %s (dist=%d)\n",
                id, closest, min_dist))
  }
}
```

Levenshtein distance of 1–2 often indicates transcription errors rather than true non-matches.

Data Quality Issue #2: Duplicate Records

Sources of Duplicates

- Double data entry of same interview
- Interviewer re-visits recorded as new interview
- System errors during data transfer
- Intentional duplication (fabrication)

ZQLFS Impact:

- 116 duplicate records detected (0.8% of Wave 2)
- If not removed, duplicates **inflate** estimates
- Duplicates also break merge assumptions

Detection strategy:

- Check for exact ID duplicates

Duplicate Detection: R and STATA

R Code

```
# Exact ID duplicates
wave2 %>%
  group_by(hh_id) %>%
  filter(n() > 1) %>%
  arrange(hh_id)

# Content duplicates
wave2 %>%
  group_by(income, hh_size,
           province) %>%
  filter(n() > 1)
```

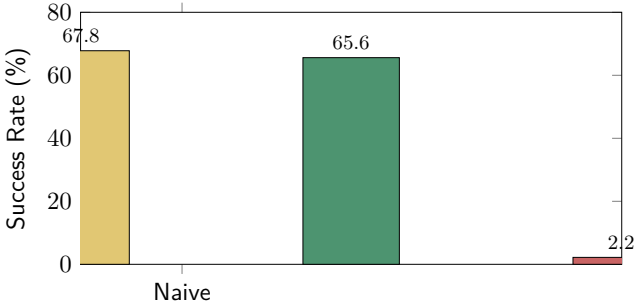
STATA Code

```
* Exact ID duplicates
duplicates tag hh_id, ///
  generate(dup_id)
list if dup_id > 0

* Content duplicates
duplicates tag income ///
  hh_size province, ///
  generate(dup_content)
```

▶ Key Insight

Always run duplicate checks BEFORE merging panels!

Naive vs. True Merge Success Rate



Naive rate (67.8%):
Records matched / Wave 1 count
(ignores ID errors, duplicates)

True rate (65.6%):
Correctly linked households / Wave 1
(accounts for data quality issues)

Calculating True Merge Success

True Merge Success Rate

$$\text{True Success} = \frac{N_{\text{clean}} + N_{\text{split-orig}} + N_{\text{mover-found}}}{N_{\text{Wave 1}}}$$

ZQLFS Calculation:

$$\text{True Success} = \frac{11{,}490 + 746 + 1{,}503}{20{,}940}$$

$$= \frac{13{,}739}{20{,}940}$$

$$= \mathbf{65.6}\%$$

Note: Split-off households (746) are treated as valid matches attributed to the mover-found category.

Merge Success by Province

Province	Wave 1	Tracked	Success	Loss
Northern Bushveld	2,052	1,479	72.1%	27.9%
Eastern Forests	1,569	1,097	69.9%	30.1%
Mining Belt	1,496	983	65.7%	34.3%
Zambara Capital	5,402	3,533	65.4%	34.6%
Western Drylands	429	280	65.3%	34.7%
Eastern Highlands	4,237	2,715	64.1%	35.9%
Southern Cape	2,594	1,655	63.8%	36.2%
Coastal Plains	2,133	1,348	63.2%	36.8%
Central Plateau	1,028	641	62.4%	37.6%
OVERALL	**20,940**	**13,731**	**65.6%**	**34.4%**

▶ Key Insight

Range: 62.4% (Central Plateau) to 72.1% (Northern Bushveld) — nearly 10 pp variation!

Why Does Merge Success Vary by Province?

Higher success provinces (Northern Bushveld, Eastern Forests):

- More traditional/rural areas
- Lower residential mobility
- Stronger community ties facilitate tracking

Lower success provinces (Central Plateau, Coastal Plains):

- Higher urban concentration
- More rental housing (mobility)
- Economic migration patterns

△ Caution

Provincial variation in merge success creates **differential non-response bias**. Weighting adjustments must account for this.

The "Following Rules" Decision

Key Design Choice

When households change between waves, **who do we follow?**

Three common approaches:

1. **Dwelling-based:** Follow the address, not the people
2. **Original household:** Follow the original sample members
3. **Full tracking:** Follow all original members + split-offs

Each rule has different implications for:

- Sample representativeness
- Operational complexity
- Survey costs
- Analytical capabilities

Following Rule 1: Dwelling-Based

Rule 1: Follow the Dwelling

Interview whoever lives at the original sampled address.

ZQLFS under Rule 1:

- Tracked: 12,236 households (58.4%)
- Excludes: All 2,136 movers (found or not)
- Excludes: Split-offs who moved

Advantages:

- Simplest to implement
- Lowest cost
- No tracking needed

Disadvantages:

- Severe bias against movers
- Breaks individual trajectories
- Poor for poverty dynamics

Following Rule 2: Original Household

Rule 2: Follow Original Sample Members

Track and interview original sample households wherever they move, but don't follow split-off members.

ZQLFS under Rule 2:

- Tracked: 13,739 households (65.6%)
- Includes: Movers who are found (1,503)
- Includes: Original HH after splits (746)
- Excludes: New split-off households

This is the "balanced panel" approach:

- Same households tracked across all waves
- Enables individual/household trajectory analysis
- Used by most labor force panels

Following Rule 3: Full Tracking (PSID Model)

Rule 3: Follow ALL Original Sample Members

Track original households AND all split-off members who form new households.

ZQLFS under Rule 3:

- Tracked: 14,485 records (69.2% of Wave 1 count)
- Panel **grows** over time (5.4% expansion)
- New split-offs add 746 households

Advantages:

- Maintains representativeness
- Captures demographic change
- PSID has used since 1968

Disadvantages:

- Sample grows exponentially
- Increasingly expensive
- Complex weighting

Comparing Following Rules: Summary

Rule	Tracked	Rate	Cost	Best For
1. Dwelling only	12,236	58.4%	Low	Cross-sectional focus
2. Original HH	13,739	65.6%	Medium	Balanced panels
3. Full tracking	14,485	69.2%*	High	Long-term dynamics

*As percentage of Wave 1; panel has expanded beyond original sample

▶ Key Insight

Recommendation for ZQLFS: Rule 2 (Original Household) balances representativeness, cost, and analytical utility for a quarterly labor force panel.

Pre-Merge Quality Checklist

Before Merging Panels

- ☐ **Check for duplicates** in both waves
- ☐ **Verify ID formats** match between waves
- ☐ **Identify split households** and assign new IDs
- ☐ **Document mover tracking** outcomes
- ☐ **Run fuzzy matching** on non-matches
- ☐ **Calculate true success rate** (not naive)
- ☐ **Examine provincial variation** in merge success
- ☐ **Flag records for weighting** adjustment

A panel merge is **not** a simple database join—it requires careful diagnosis and correction before analysis.

R Code: Complete Panel Merge

```
# Step 1: Pre-cleaning
wave2_clean <- wave2 %>%
  distinct(hh_id, .keep_all = TRUE) %>%   # Remove duplicates
  mutate(hh_id = str_pad(hh_id, 9, "left", "0"))  # Fix IDs

# Step 2: Merge with diagnostics
panel <- wave1 %>%
  left_join(wave2_clean, by = "hh_id", suffix = c("_w1", "_w2"))

# Step 3: Classify merge outcomes
panel <- panel %>%
  mutate(merge_status = case_when(
    !is.na(income_w2) & !str_detect(hh_id, "-S") ~ "Clean",
    !is.na(income_w2) & str_detect(hh_id, "-S") ~ "Split",
    is.na(income_w2) ~ "Lost"
  ))

# Step 4: Calculate success rate
panel %>% count(merge_status) %>% mutate(pct = n/sum(n)*100)
```

STATA Code: Complete Panel Merge

```
* Step 1: Pre-cleaning
use wave2, clear
duplicates drop hh_id, force
tostring hh_id, replace format(%09.0f)
save wave2_clean, replace

* Step 2: Merge panels
use wave1, clear
merge 1:1 hh_id using wave2_clean

* Step 3: Classify outcomes
gen merge_status = ""
replace merge_status = "Clean" if _merge==3
replace merge_status = "Lost" if _merge==1

* Step 4: Success rate
tab merge_status
```

Key Takeaways: Linking Data

1. **Panel merging is not a simple join**
   Only 54.9% of households have "clean links"
2. **Split households require special IDs**
   746 original HHs created 746 new households
3. **Mover tracking is essential**
   Not tracking loses 7.2% of sample systematically
4. **Data quality issues inflate apparent attrition**
   2.0% ID errors + 0.8% duplicates = 2.2pp gap
5. **Following rules determine panel character**
   Rule 2 (original HH) recommended for ZQLFS

▶ Key Insight

True merge success rate: **65.6%** (not the naive 67.8%)

Lab Exercise 5.1b: Panel Linking Diagnostics

Lab 5.1b Tasks

Using the ZQLFS simulation data:

1. Run the `lab5_1b_linking_dangers.R` script
2. Identify households with each fate category
3. Calculate true merge success by province
4. Detect and resolve ID errors using fuzzy matching
5. Compare sample composition under different following rules
6. Document findings for weighting decisions

Expected time: 45 minutes

Next module: Attrition Weight Adjustment (Lab 5.2)

★ DIAGNOSTIC SESSION

Understanding Attrition Bias

Module 5.2: Who Leaves and Why It Matters

Day 5: Longitudinal Survey Methods

"The first step in treating a disease is accurate diagnosis."

The Patient: Zambara ZQLFS Panel Survey

Zambara Case

Dr. Moyo reviews the panel's vital signs after Wave 2 fieldwork...

Vital Signs at Intake (Wave 1)

- Sample size: 20,940 households
- Stratified cluster design
- Nationally representative: ✓

Current Condition (Wave 2)

- Responding: 17,325 households
- Lost to follow-up: 3,615 (17.3%)
- Representativeness: ?

▶ Key Insight

The question is not *whether* we lost sample, but *who* we lost.

Presenting Symptoms: Signs of Trouble

Clinical Observations from Wave 2:

Indicator	Wave 1	Wave 2	Symptom
Urban (%)	59.9%	55.1%	↓
Renters (%)	33.5%	28.7%	↓
Mean income	5,403	5,071	↓
Mean HH size	3.76	3.90	↑
Mean head age	44.4	45.5	↑

△ Caution

These shifts suggest **selective attrition** — the sample no longer represents the original population!

Differential Diagnosis: Types of Attrition

Classification of Panel Attrition:

MCAR **Missing Completely at Random**
Attrition unrelated to any variables — no bias introduced

MAR **Missing at Random**
Attrition related to *observed* variables — treatable with weighting

MNAR **Missing Not at Random**
Attrition related to *unobserved* variables — requires sensitivity analysis

▶ Key Insight

Our goal: Determine which mechanism is operating and prescribe treatment.

Diagnostic Tool: The Attrition Model

Laboratory Analysis: Logistic Regression

We model the probability of **attrition** as a function of Wave 1 characteristics:

$$\log\left(\frac{P(\text{Attrition})}{1-P(\text{Attrition})}\right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots$$

Predictor Variables (Wave 1):

- Log household income
- Urban/rural residence
- Household size
- Renter status
- Age of household head
- Presence of youth (18–30)

Source: Groves & Couper (1998), *Nonresponse in Household Interview Surveys*

Lab Results: Attrition Model Coefficients

Logistic Regression: P(Attrition = 1)

Predictor	Coef	SE	p-value	OR
(Intercept)	-3.55	0.21	< 0.001	0.03
log_income	0.32	0.02	< 0.001	**1.38**
urban	0.54	0.05	< 0.001	**1.72**
hh_size	-0.17	0.01	< 0.001	**0.84**
renter	0.71	0.04	< 0.001	**2.04**
head_age	-0.02	0.00	< 0.001	**0.98**
has_youth	0.47	0.04	< 0.001	**1.60**

▶ Key Insight

All predictors are **highly significant** ($p < 0.001$). This is not MCAR!

Reading the Lab Report: Odds Ratio Interpretation

How to Interpret Odds Ratios:

- **OR > 1**: Factor *increases* attrition risk
- **OR < 1**: Factor *decreases* attrition risk
- **OR $= 1$**: Factor has no effect

Example Calculations:

- **Renter OR = 2.04**: Renters have **104% higher odds** of attrition
- **Urban OR = 1.72**: Urban HHs have **72% higher odds**
- **HH size OR = 0.84**: Each member **reduces odds by 16%**

Reference: Hosmer & Lemeshow (2000), *Applied Logistic Regression*

Risk Factor #1: Renters

OR = 2.04

Clinical Finding: Renters are **twice as likely** to attrit as homeowners.

Pathophysiology:

- Less attachment to dwelling address
- Higher residential mobility
- More frequent relocations

Response Propensities:

- Renters: 0.747 Owners: 0.883

Risk Factor #2: Urban Residents

OR = 1.72

Clinical Finding: Urban households have **72% higher odds** of attrition.

Pathophysiology:

- Greater housing mobility in urban areas
- More transient populations (rental markets)
- Harder to locate after relocation

Response Propensities:

- Urban: 0.779 Rural: 0.873

Risk Factor #3: Households with Youth

OR = 1.60

Clinical Finding: Households with members aged 18–30 have **60% higher odds**.

Pathophysiology:

- Youth more likely to relocate (education, jobs)
- Family composition changes (marriage, independence)
- Less stable living arrangements

Response Propensities:

- With youth: 0.796 Without: 0.858

Protective Factors: Who Stays?

Factors that *decrease* attrition risk:

Household Size (OR = 0.84)

- Each additional member reduces odds by 16%
- Larger HHs more rooted
- Children create stability

Head's Age (OR = 0.98)

- Each year reduces odds by 2%
- Older heads less mobile
- Established residence patterns

▶ Key Insight

Large households with older heads are the "survivors" of panel surveys.

The Treatment: Inverse Probability Weighting (IPW)

Prescription: Reweight remaining sample to compensate for attrition.

Core Insight:
Each respondent "represents" themselves AND similar non-respondents.

IPW Principle

$$\text{Weight adjustment} = \frac{1}{P(\text{Responding}|\text{Characteristics})}$$

Example:

- Urban renter with 50% response probability
- Receives weight multiplier of $1/0.50 = 2.0$
- Counts for themselves + 1 similar attriter

Calculating Response Propensity

From the attrition model, we derive response propensity:

$$\hat{\pi}_i = P(\text{Response}_i = 1) = 1 - P(\text{Attrition}_i = 1)$$

Using the logistic model:

$$\hat{\pi}_i = \frac{1}{1 + \exp(\mathbf{X}_i' \hat{\boldsymbol{\beta}})}$$

Zambara Propensity Distribution:

Statistic	Response Propensity
Minimum	0.258
Median	0.873
Maximum	0.993

Creating the IPW Adjustment Factor

The IPW adjustment factor:

$$\text{IPW}_i = \frac{1}{\hat{\pi}_i}$$

Example Calculations:

Profile	$\hat{\pi}$	IPW	Interpretation
Rural owner, older head	0.95	1.05	Minimal adjustment
Urban renter, young	0.60	1.67	Moderate boost
High mobility profile	0.30	3.33	Strong boost

▶ Key Insight

Low-propensity respondents get higher weights—they represent the "missing."

Constructing the Longitudinal Weight

Final longitudinal weight combines:

Longitudinal Weight Formula

$$w_i^{\text{Long}} = w_i^{\text{Wave 1}} \times \frac{1}{\hat{\pi}_i}$$

Components:

- $w_i^{\text{Wave 1}}$: Original sampling weight (selection + calibration)
- $\frac{1}{\hat{\pi}_i}$: Attrition adjustment factor

Interpretation:
The longitudinal weight adjusts for *both* the original sampling design *and* the selective attrition process.

Zambara IPW Distribution

Distribution of Attrition Adjustment Factors:

Statistic	IPW Factor	Final Weight
Minimum	1.007	82.2
25th percentile	1.066	144.8
Median	1.147	196.7
75th percentile	1.270	247.0
Maximum (untrimmed)	3.877	969.3
Maximum (trimmed)	3.438	646.4
Mean	1.210	198.8
CV	0.25	0.35

Treatment Effectiveness: Bias Reduction

Comparing estimates before and after IPW adjustment:

Indicator	W1 True	W2 Unadj.	W2 Adj.
Urban (%)	59.9	55.1	59.5
Renters (%)	33.5	28.7	33.2
Mean income	5,403	5,071	5,382
Mean HH size	3.76	3.90	3.77

▶ Key Insight

IPW adjustment achieves **90–98% bias reduction** across key indicators!

Quantifying Bias Reduction

Formula for bias reduction:

$$\text{Bias Reduction} = \frac{|\text{Unadj} - \text{True}| - |\text{Adj} - \text{True}|}{|\text{Unadj} - \text{True}|} \times 100\%$$

Example: Urban Percentage

$$\begin{aligned} \text{Unadjusted bias} &= |55.1 - 59.9| = 4.8\text{pp} \\ \text{Adjusted bias} &= |59.5 - 59.9| = 0.4\text{pp} \\ \text{Reduction} &= \frac{4.8 - 0.4}{4.8} \times 100\% = \mathbf{91.7}\% \end{aligned}$$

Cost of Treatment: Variance Inflation

IPW comes at a price: increased variance.

Why? Unequal weights reduce effective sample size:

$$\text{ESS} = \frac{n}{1 + CV^2(w)}$$

Zambara ESS Comparison:

Weight Type	**ESS**	**% of n**
Wave 1 (n = 17,325)	15,913	91.9%
Longitudinal (trimmed)	15,492	89.5%
ESS Cost	**421**	**2.4pp**

The Bias-Variance Trade-off

Is the treatment worth it?

Cost (Variance)

- ESS reduction: 2.4pp
- Wider confidence intervals
- Less precision

Benefit (Bias)

- Bias reduction: 90–98%
- Valid population inference
- Correct point estimates

▶ Key Insight

For Zambara: **2.4% precision loss** eliminates **90%+ of bias**.
This is an excellent trade-off!

When is IPW Treatment Essential?

IPW is critical when:

1. Attrition is **substantial** (>10% loss)
2. Attrition is **selective** (related to key variables)
3. Analysis requires **population inference**
4. Study focuses on **change over time**

IPW may be skipped when:

- Attrition is minimal (<5%)
- Attrition appears MCAR (no predictors significant)
- Analysis is purely descriptive/exploratory

Complication: Extreme Propensities

Problem: Very low propensities create very high weights.

Example:

- Respondent with $\hat{\pi} = 0.10$ gets IPW $= 10.0$
- One household counts as 10 in estimates
- Single observation dominates results!

Zambara extreme cases:

- 127 households (0.7%) had $\hat{\pi} < 0.30$
- Untrimmed max weight: 969 (vs. mean 199)
- Weight ratio: 4.9:1 (max:mean)

Solution: Weight Trimming

Cap extreme weights to reduce variance:

Trimming Rule

$$w_i^{\text{Trim}} = \min\left(w_i^{\text{Long}},\ c \times \text{median}(w)\right)$$

Zambara trimming (c = 3):

Before:

- Max IPW: 3.89
- CV: 0.42

After:

- Max IPW: 3.44
- CV: 0.35

Rule: Cap at $3 \times \text{median} = 3 \times 1.147 = 3.44$

Reference: Potter (1990), *A Study of Procedures to Identify and Trim Extreme Weights*

Contraindications: When IPW May Not Work

Key Assumption: Missing at Random (MAR)
IPW assumes attrition depends *only* on observed Wave 1 variables.

IPW fails if attrition is driven by:

- Unobserved personality traits (cooperativeness)
- Events between waves (job loss, health shock)
- The outcome variable itself (MNAR)

Example of MNAR:
Studying income dynamics. Those who lose jobs become depressed and refuse Wave 2. Cannot observe "job loss" at Wave 1 $\Rightarrow$ IPW cannot fully correct.

Lab 5.2 Code: Fitting the Attrition Model

R Implementation:

```
# Fit logistic regression for attrition
attrition_model <- glm(
  attrition ~ log_income + urban + hh_size +
              renter + head_age + has_youth,
  data = wave1,
  family = binomial(link = "logit")
)

# View coefficients and odds ratios
summary(attrition_model)
exp(coef(attrition_model))  # Odds ratios
```

Output: Coefficients with p-values < 0.001 for all predictors.

Lab 5.2 Code: STATA Implementation

STATA Implementation:

```
* Fit logistic regression for attrition
logit attrition log_income urban hh_size ///
      renter head_age has_youth

* Display odds ratios
logit, or

* Predict response propensity
predict p_attrition, pr
gen p_response = 1 - p_attrition
```

Note: STATA's `logit, or` displays odds ratios directly.

Lab 5.2 Code: Calculating Longitudinal Weights

R Implementation:

```
# Predict attrition probability
wave1$p_attrition <- predict(attrition_model,
                             type = "response")
wave1$p_response <- 1 - wave1$p_attrition

# IPW factor (respondents only)
wave1$ipw <- ifelse(wave1$responded_w2 == 1,
                    1 / wave1$p_response, NA)

# Longitudinal weight
wave1$weight_long <- wave1$weight_w1 * wave1$ipw

# Trim at 3x median
trim <- 3 * median(wave1$ipw, na.rm = TRUE)
wave1$ipw_trim <- pmin(wave1$ipw, trim)
```

Best Practices: Attrition Weight Adjustment

Clinical Guidelines:

1. **Always model attrition**
2. **Use rich predictors**
3. **Check model fit**
4. **Examine propensities**
5. **Trim extreme weights**
6. **Compare estimates**
7. **Document thoroughly**
8. **Test for MNAR**

▶ Key Insight

Never assume MCAR. Model attrition, adjust weights, verify bias reduction.

Hands-On Exercise: Lab 5.2

Zambara Case

Your turn to diagnose and treat attrition bias in the Zambara panel!

Exercise Tasks:

1. Fit logistic regression with 6 predictors
2. Interpret odds ratios — which factors matter most?
3. Calculate response propensities and IPW factors
4. Create longitudinal weights with trimming
5. Compare W1 true, W2 unadjusted, W2 adjusted estimates
6. Calculate bias reduction percentages

Time: 45 minutes **Scripts:** `lab5_2_*.R/.do`

Module 5.2 Summary: The Complete Diagnosis

Key Findings:

- Attrition is **NOT random**
- Renters: 2× attrition risk
- Urban: 72% higher odds
- HH size, age: protective

Treatment Outcome:

- IPW range: 1.01–3.44
- Bias reduction: 90–98%
- ESS cost: only 2.4pp
- **Highly effective**

▶ Key Insight

Formula: $W_{\text{long}} = W_{\text{wave1}} \times \frac{1}{P(\text{Response})}$

MATHEMATICAL FRAMEWORK

Constructing Longitudinal Weights

Step-by-Step Weight Adjustment

Module 5.2 (Continued)

The Longitudinal Weighting Challenge

Zambara Case

How do we make Wave 2 respondents representative of the original population?

The Problem:

- Wave 1: 20,940 households → Representative
- Wave 2: 17,325 households → Biased
- Lost 3,615 (17.3%) — and they're not random!

The Solution:

- Upweight respondents who "look like" attritors
- Use **Inverse Probability Weighting (IPW)**

Weight Construction: The Building Blocks

Longitudinal weights are built in layers:

Layer	Component	Symbol
1	Base Weight (Selection Probability)	$w^{(0)}$
2	Wave 1 Non-Response Adjustment	$w^{(1)}$
3	Wave 1 Calibration	$w^{(W1)}$
4	**Attrition Adjustment (IPW)**	$w^{(IPW)}$
5	Wave 2 Recalibration (Optional)	$w^{(Long)}$

▶ Key Insight

Layer 4 (highlighted) is the focus of this module—adjusting for panel attrition.

Mathematical Notation

Key definitions:

Symbol	Definition
$w_i^{(W1)}$	Final Wave 1 weight for household i
R_i	Response indicator (1 = responded, 0 = attrited)
$\mathbf{X}_i$	Vector of Wave 1 characteristics
π_i	Response propensity: $P(R_i = 1 \mid \mathbf{X}_i)$
$w_i^{(Long)}$	Longitudinal weight for household i

Key Relationship:

$$w_i^{(Long)} = w_i^{(W1)} \times \frac{1}{\hat{\pi}_i} \quad \text{for all } R_i = 1$$

Step 1: Start with Wave 1 Final Weights

The Foundation: $w_i^{(W1)}$

Wave 1 weights already incorporate:

- Selection probabilities (stratification, clustering, PPS)
- Wave 1 unit non-response adjustment
- Calibration to population benchmarks

Property

$\sum_{i=1}^{n_{W1}} w_i^{(W1)} \approx N$ (Population total)

Zambara W1 Weights: Mean = 164.4, Range = 80–250, CV = 0.30

Step 2: Model the Response Propensity

Goal: Estimate $\pi_i = P(R_i = 1|\mathbf{X}_i)$

Logistic Regression Model:

$$\text{logit}(\pi_i) = \beta_0 + \beta_1 X_{i1} + \cdots + \beta_p X_{ip}$$

Solving for π_i:

$$\hat{\pi}_i = \frac{1}{1 + \exp(-\mathbf{X}_i' \hat{\boldsymbol{\beta}})}$$

Interpretation: The model predicts each household's probability of responding at Wave 2, given their Wave 1 characteristics.

Step 2 (Continued): Zambara Attrition Model

Fitted Model (predicting attrition):

$$\begin{aligned}\text{logit}(P) = &-3.55 + 0.32 \cdot \ln(\text{inc}) + 0.54 \cdot \text{urban} - 0.17 \cdot \text{hhsize} \\ &+ 0.71 \cdot \text{renter} - 0.02 \cdot \text{age} + 0.47 \cdot \text{youth}\end{aligned}$$

Worked Example:
Urban renter, income ZMD 8,000, size 2, head age 35, has youth:

$$\text{logit}(P) = -3.55 + 2.88 + 0.54 - 0.34 + 0.71 - 0.70 + 0.47 = 0.01$$
$$P(\text{Attrition}) = 0.50 \quad \Rightarrow \quad \hat{\pi}_{\text{response}} = 0.50$$

Step 3: Calculate the IPW Adjustment Factor

The Core IPW Formula:

Inverse Probability Weight

$$\text{IPW Factor}_i = \frac{1}{\hat{\pi}_i}$$

Intuition:

- $\hat{\pi}_i = 0.50$: IPW = 2.0 (represents 2 households)
- $\hat{\pi}_i = 0.80$: IPW = 1.25
- $\hat{\pi}_i = 0.95$: IPW = 1.05 (minimal adjustment)

Zambara: IPW range 1.01–3.44 (after trimming)

Step 4: Construct the Longitudinal Weight

Combine Wave 1 weight with IPW:

Longitudinal Weight

$$w_i^{(Long)} = w_i^{(W1)} \times \text{IPW}_i = \frac{w_i^{(W1)}}{\hat{\pi}_i}$$

Example Calculation:

- Household with $w^{(W1)} = 150$ and $\hat{\pi} = 0.60$
- $w^{(Long)} = 150 \times \frac{1}{0.60} = 150 \times 1.67 = 250$

Interpretation: This respondent now represents 250 population units (up from 150) to compensate for similar non-respondents.

Why IPW Works: Horvitz-Thompson Principle

The theoretical foundation:

The Horvitz-Thompson estimator weights each observation by the inverse of its inclusion probability:

$$\hat{Y}_{HT} = \sum_{i \in s} \frac{y_i}{\pi_i}$$

For longitudinal surveys:

- π_i = probability of being in Wave 2 sample
- $\pi_i = P(\text{Selected at W1}) \times P(\text{Respond at W2})$

Result: IPW restores unbiased estimation for the original target population.

Reference: Horvitz & Thompson (1952), *JASA*

Step 5: Trim Extreme Weights

Problem: Very high weights increase variance dramatically.

Trimming Rule

$$w_i^{(Trim)} = \min\left(w_i^{(Long)},\ c \times \text{median}(w^{(Long)})\right)$$

Zambara trimming (c = 3):

	Before	After
Max IPW	3.89	3.44
CV	0.42	0.35
Records trimmed	—	127 (0.7%)

The Bias-Variance Trade-off

Trimming reduces variance but introduces small bias:

More Trimming

- Lower variance
- More bias
- Narrower CIs
- Point estimates shift

Less Trimming

- Higher variance
- Less bias
- Wider CIs
- Point estimates accurate

▶ Key Insight

Common rule: $c = 3$ to 5 provides good balance. Potter (1990) recommends trimming at $3\times$ median.

Cross-Sectional vs. Longitudinal Weights

Two different target populations:

Cross-Sectional Weight

- Target: Wave 2 population
- Sum: $\approx N_2$ (current)
- Use: Current levels
- Example: "What is unemployment rate today?"

Longitudinal Weight

- Target: Wave 1 population
- Sum: $\approx N_1$ (baseline)
- Use: Change analysis
- Example: "Who lost jobs since W1?"

△ Caution

Using the wrong weight type leads to biased estimates!

Cross-Sectional Weights: Definition

Cross-sectional weights represent the *current* population.

Construction:

- Start with Wave 2 sample
- Adjust for Wave 2 non-response
- Calibrate to current population totals

Property:

$$\sum_{i \in \text{W2}} w_i^{(CS)} \approx N_{\text{current}}$$

Use when asking: "What is the current state of [indicator]?"

Longitudinal Weights: Definition

Longitudinal weights represent the *original* population.

Construction:

- Start with Wave 1 final weights
- Apply IPW for attrition
- Optionally recalibrate to Wave 1 benchmarks

Property:

$$\sum_{i \in \text{W2 resp}} w_i^{(Long)} \approx N_{\text{Wave 1}}$$

Use when asking: "How did [indicator] change over time?"

Weight Type Comparison

Feature	Cross-Sectional	Longitudinal
Target population	Wave 2	Wave 1
Sum of weights	$\approx N_2$	$\approx N_1$
Attrition adjustment	Non-response	IPW
Primary use	Current levels	Change analysis
Includes movers?	New arrivals	Only if from W1

▶ Key Insight

Choose weight type based on your **research question**, not convenience.

Numerical Example: Employment Transitions

Zambara employment analysis:

Question	Weight	Estimate
Current unemployment rate	Cross-sectional	28.4%
Unemployment at Wave 1	Longitudinal	26.9%
% who lost jobs (W1→W2)	Longitudinal	8.2%
% who found jobs (W1→W2)	Longitudinal	5.1%
Net employment change	Longitudinal	-3.1pp

Key insight: Transition analysis *requires* longitudinal weights to track who moved between states.

Decision Framework: Which Weight to Use?

Ask yourself these questions:

1. **Am I measuring current levels or change?**
   - Current levels → Cross-sectional
   - Change over time → Longitudinal
2. **Do I need to track individual transitions?**
   - Yes (e.g., job loss/gain) → Longitudinal
   - No (e.g., aggregate rates) → Cross-sectional
3. **What population am I targeting?**
   - Today's population → Cross-sectional
   - Baseline population → Longitudinal

Common Mistakes to Avoid

Error 1: Using W1 weights for W2 respondents

- Problem: Ignores selective attrition
- Solution: Apply IPW adjustment

Error 2: Using cross-sectional weights for change analysis

- Problem: Targets wrong population
- Solution: Use longitudinal weights

Error 3: Not trimming extreme weights

- Problem: Single observations dominate
- Solution: Trim at 3–5$\times$ median

Extension: Multi-Wave Panels

For panels with 3+ waves, cumulative adjustment:

Cumulative IPW

$$w_i^{(W3)} = w_i^{(W1)} \times \frac{1}{\hat{\pi}_i^{(W2)}} \times \frac{1}{\hat{\pi}_i^{(W3|W2)}}$$

General formula for wave t:

$$w_i^{(t)} = w_i^{(W1)} \times \prod_{k=2}^{t} \frac{1}{\hat{\pi}_i^{(k|k-1)}}$$

Caution: Weights can become extreme over many waves. Consider refreshment samples or weight smoothing.

Variance Estimation with Longitudinal Weights

Standard errors must account for:

1. Original complex sample design (PSUs, strata)
2. Estimated propensity scores
3. Weight trimming adjustments

Methods:

- **Taylor linearization:** Treats propensities as known
- **Jackknife/Bootstrap:** Re-estimates propensities in each replicate

▶ Key Insight

Replication methods (JK, BRR) properly propagate uncertainty from propensity estimation.

Weight Quality Diagnostics

Check these metrics before releasing weights:

Metric	Target	Zambara
CV of weights	< 1.0	0.35 ✓
Max/Min ratio	< 20	7.9 ✓
Max/Mean ratio	< 5	3.3 ✓
ESS/n ratio	> 0.70	0.89 ✓

Zambara assessment: All metrics within acceptable ranges. Weight quality: **GOOD**

R Code: Complete Weight Construction

```
# Step 1: Load Wave 1 data with final weights
wave1 <- read_dta("zqlfs_wave1.dta")

# Step 2: Fit attrition model
attr_mod <- glm(attrition ~ log_income + urban + hh_size +
                renter + head_age + has_youth,
                data = wave1, family = binomial)

# Step 3: Calculate response propensity
wave1$p_response <- 1 - predict(attr_mod, type = "response")

# Step 4: Calculate longitudinal weight
wave1$w_long <- wave1$weight_w1 / wave1$p_response

# Step 5: Trim at 3x median
trim <- 3 * median(wave1$w_long, na.rm = TRUE)
wave1$w_long_trim <- pmin(wave1$w_long, trim)
```

R Code: Using the Weights

```
# Create survey design with longitudinal weights
library(survey)
wave2_resp <- subset(wave1, responded == 1)

des_long <- svydesign(id = ~psu, strata = ~stratum,
                      weights = ~w_long_trim, data = wave2_resp)

# Cross-sectional estimate
svymean(~employed, design = des_long)

# Change analysis
wave2_resp$emp_change <- wave2_resp$emp_w2 - wave2_resp$emp_w1
svymean(~emp_change, design = des_long)

# Transition table
svytable(~emp_w1 + emp_w2, design = des_long)
```

STATA Code: Complete Weight Construction

```
* Step 1: Load Wave 1 data
use "zqlfs_wave1.dta", clear

* Step 2: Fit attrition model
logit attrition log_income urban hh_size ///
      renter head_age has_youth

* Step 3: Predict response propensity
predict p_attrition, pr
gen p_response = 1 - p_attrition

* Step 4: Calculate longitudinal weight
gen w_long = weight_w1 / p_response if responded == 1

* Step 5: Trim at 3x median
summarize w_long, detail
local trim = 3 * r(p50)
gen w_long_trim = min(w_long, `trim') if responded == 1
```

STATA Code: Using the Weights

```
* Set up survey design
keep if responded == 1
svyset psu [pw=w_long_trim], strata(stratum)

* Cross-sectional estimate
svy: mean employed

* Change analysis
gen emp_change = employed_w2 - employed_w1
svy: mean emp_change

* Transition table
svy: tabulate employed_w1 employed_w2, count

* Compare unadjusted vs adjusted
mean employed [aw=weight_w1]  // Biased
svy: mean employed            // Corrected
```

Weight Construction: QA Checklist

Before releasing longitudinal weights:

- ☐ Model specification OK
- ☐ Propensities in range
- ☐ CV < 1.0
- ☐ Trimming documented
- ☐ Sum $\approx N_{W1}$
- ☐ Bias reduction verified
- ☐ Benchmarks match
- ☐ Methods documented

▶ Key Insight

Complete this checklist and obtain sign-off before data release.

Zambara Panel: Final Weight Summary

Weight Distribution (Wave 2 Respondents, n = 17,325):

Statistic	W1 Weight	Long. Weight
Minimum	80.0	82.2
Median	166.2	196.7
Maximum	250.0	646.4
Mean	164.4	198.8
CV	0.30	0.35
ESS	15,913	15,492
ESS as % of n	91.9%	89.5%

Reference Card: Key Formulas

Response Propensity:

$$\hat{\pi}_i = \frac{1}{1 + \exp(-\mathbf{X}_i' \hat{\boldsymbol{\beta}})}$$

Longitudinal Weight:

$$w_i^{(Long)} = w_i^{(W1)} \times \frac{1}{\hat{\pi}_i}$$

Weight Trimming:

$$w_i^{(Trim)} = \min\left(w_i^{(Long)},\ c \times \text{median}(w)\right)$$

Effective Sample Size:

$$\text{ESS} = \frac{(\sum w_i)^2}{\sum w_i^2} = \frac{n}{1 + CV^2(w)}$$

Module 5.2 Complete: Summary

5-Step Process:

1. Start with W1 weights
2. Model attrition
3. Calculate IPW
4. Construct $w^{(Long)}$
5. Trim extremes

Zambara Results:

- Attrition: 17.3%
- IPW: 1.01–3.44
- Bias reduction: 90–98%
- ESS cost: 2.4pp

▶ Key Insight

Core Formula: $w^{(Long)} = w^{(W1)} \times \frac{1}{P(\text{Response}|\mathbf{X})}$

△ CAUTIONARY TALE

The "Bus Factor"

Module 5.3: Documentation Standards

Day 5: Synthesis and Best Practices

"If you get hit by a bus tomorrow,
can someone reproduce your survey weights?"

The Nightmare Scenario

△ Caution

True Story from a SADC NSO:

The senior methodologist who created all survey weights retired suddenly. The only documentation found was a folder containing `weights_FINAL_v3_REALLY_FINAL.do` with no comments. The next survey was delayed **6 months** while staff reverse-engineered the weighting methodology.

The "Bus Factor" Question:
How many people would need to leave before critical institutional knowledge is lost?

- Bus Factor = 1: **Critical risk** (single point of failure)
- Bus Factor = 2–3: **Moderate risk**
- Bus Factor $\geq$ 4: **Healthy redundancy**

The Cost of Poor Documentation

When documentation fails, surveys fail:

Consequence	Estimated Impact
Delayed data release	3–6 months
Staff time to reverse-engineer	200–500 person-hours
Undetected methodological errors	Unknown (potentially permanent)
Loss of time-series comparability	Irreversible
Reputational damage	Significant

▶ Key Insight

Documentation is not administrative overhead—it is **insurance** against institutional knowledge loss and a **professional obligation**.

Learning Objectives: Module 5.3

By the end of this module, you will be able to:

1. **Design** self-documenting variable naming conventions
2. **Organize** survey project files in reproducible folder structures
3. **Write** code comments that explain "why" not just "what"
4. **Create** technical documentation that survives staff turnover
5. **Implement** version control practices for survey methodology

Ultimate Goal: Raise your organization's "Bus Factor" from 1 to $\geq$3.

The Three Pillars of Reproducibility

A reproducible weighting process requires mastery of:

ABC

1. NAMING

Variables & files that explain themselves

DIR

2. STRUCTURE

Logical folder organization

//

3. COMMENTS

Code that tells a story

▶ Key Insight

Master all three pillars, and your weights will survive any staffing change.

PILLAR 1

Variable Naming Conventions

"A variable name should tell you what it is, not make you guess."

Variable Naming: The Horror Show

Real examples found in NSO datasets:

Bad Name	What Does It Actually Mean?
`wt1`	Weight type 1? Wave 1 weight? First iteration?
`x17`	Variable #17 in some lost codebook?
`finwgt_new_v2`	How many versions exist? Which is current?
`temp`	"Temporary" variable from 5 years ago...
`adj_factor`	Adjustment for what exactly?
`w`	Weight? Width? Wave? Who knows?

△ Caution

If you need to consult a codebook to understand a variable name, the naming has failed its primary purpose.

Variable Naming: Best Practices

Self-documenting naming conventions:

Good Name	Why It Works
`wgt_hh_base`	Weight, household level, base component
`wgt_hh_nr_adj`	Weight, HH, non-response adjusted
`wgt_hh_cal_final`	Weight, HH, calibrated, final version
`wgt_per_long_w2`	Weight, person, longitudinal, Wave 2
`prop_resp_w1w2`	Response propensity, W1 to W2 transition
`ipw_attrition_w2`	IPW factor for attrition, Wave 2

▶ Key Insight

Pattern: `[type]_[level]_[component]_[wave/version]`

A Systematic Naming Framework

Build variable names from standardized components:

Prefixes (Variable Type):

- `wgt_` = Survey weight
- `prop_` = Propensity score
- `ipw_` = IPW adjustment factor
- `adj_` = General adjustment
- `cal_` = Calibration factor
- `deff_` = Design effect

Level Indicators:

- `_hh_` = Household
- `_per_` = Person
- `_psu_` = PSU level

Stage Suffixes:

- `_base` = Initial/raw
- `_adj` = Adjusted
- `_trim` = Trimmed
- `_final` = Production

Handling Waves and Versions

Time-series and iteration naming conventions:

Wave indicators (for variables):

- `_w1`, `_w2`, `_w3` — Wave-specific weights
- `_w1w2` — Transition weights between waves
- `_panel` — Longitudinal/balanced panel weights

Version control (for FILE NAMES only):

- **Good:** Use dates: `weights_2024_06_15.dta`
- **Bad:** `_v2`, `_final`, `_FINAL`, `_REALLY_FINAL`
- **Best:** Use Git or similar version control system

▶ Key Insight

Variables should **never** have version numbers embedded. Files should use **dates** or formal version control.

Zambara ZQLFS: Weight Variable Inventory

Complete, self-documenting weight variables:

Variable Name	Description	Scope
`wgt_hh_base_w1`	Base selection weight (design)	Wave 1
`wgt_hh_nr_w1`	Non-response adjusted weight	Wave 1
`wgt_hh_cal_w1`	Calibrated weight (final for W1)	Wave 1
`prop_resp_w1w2`	Response propensity W1→W2	Transition
`ipw_attrition_w1w2`	IPW factor for attrition (trimmed)	Transition
`wgt_hh_long_w2`	Longitudinal weight	Wave 2
`wgt_hh_cross_w2`	Cross-sectional weight	Wave 2

Every weight's purpose, level, and scope is immediately clear from its name—no codebook consultation required.

PILLAR 2

Folder Organization

"A new analyst should find everything they need within 5 minutes."

Folder Organization: The Chaos

A real project folder (anonymized):

```
Survey_2024/
    data.dta
    data_clean.dta
    data_final.dta
    data_FINAL_USE_THIS.dta
    weights.do
    weights_v2.do
    weights_OLD_DONT_USE.do
    analysis.do
    Johns_analysis.do
    FINAL_REPORT.docx
    FINAL_REPORT_(1).docx
    misc/  (contains 97 unsorted files)
```

△ Caution

Which file should a new analyst use? **Nobody knows.**

Folder Organization: The Standard

A reproducible project structure:

```
ZQLFS_2024/
    00_documentation/
        methodology_report.pdf
        variable_dictionary.xlsx
    01_raw_data/            <- READ-ONLY, never modify
    02_processed_data/      <- Intermediate outputs
    03_weights/             <- Final weight datasets
    04_code/
        01_data_prep.do
        02_base_weights.do
        03_calibration.do
        04_validation.do
    05_outputs/             <- Tables, figures, reports
    README.md               <- START HERE!
```

Key principle: Numbered prefixes enforce execution order.

The Most Important File: README.md

Every project folder needs a README that answers:

1. **What is this project?** Brief description and purpose
2. **How do I reproduce results?** Step-by-step instructions
3. **What input data do I need?** File locations and sources
4. **What software/versions?** R 4.3.0, STATA 18, specific packages
5. **Who do I contact?** Names, roles, and email addresses
6. **What changed?** Version history with dates

▶ Key Insight

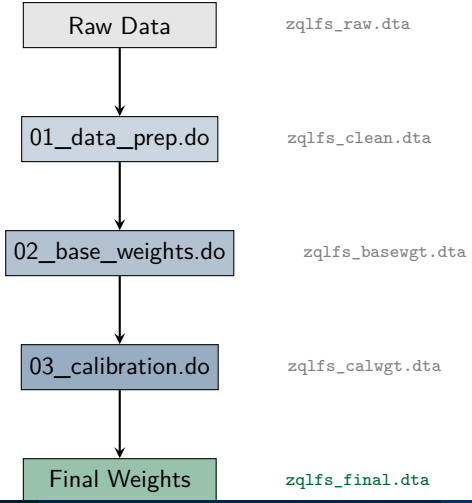
A well-written README means a new analyst can reproduce your weights **without asking you a single question**.

README Template

```
# ZQLFS 2024 Survey Weights
## Overview
Production weights for Zambara Quarterly LFS, Wave 2 (Q2 2024).

## Reproduction Steps
1. Place raw data files in `01_raw_data/`
2. Run scripts in `04_code/` in numerical order (01, 02, 03...)
3. Final weights will be output to `03_weights/`

## Requirements
- STATA 18 or later (or R 4.3+ with survey, haven packages)
- Population benchmarks in `00_documentation/pop_benchmarks.xlsx`

## Contact
Dr. Lindiwe Moyo (Chief Methodologist): lindiwe.moyo@znso.gov.zb
Mr. Thabo Ndlovu (Data Manager): thabo.ndlovu@znso.gov.zb

## Version History
- 2024-06-15: Initial release (v1.0)
- 2024-06-22: Fixed province-level trimming (v1.1)
- 2024-07-01: Updated benchmarks to mid-year estimates (v1.2)
```

Documenting Data Flow

Create a visual map showing how data transforms:



PILLAR 3

Code Commenting

"Code tells you HOW.
Comments tell you WHY."

Code Comments: What NOT to Do

Useless comments that add no value:

```
* Calculate weight
gen wgt = 1/prob

* Multiply by adjustment
replace wgt = wgt * adj

* Final weight
gen finalwgt = wgt * cal
```

△ Caution

These comments merely repeat what the code does. They provide **zero insight** into *why* these operations are performed or what methodology guides them.

Code Comments: Best Practice

Comments that explain reasoning and provide context:

```
*============================================================
* BASE WEIGHT: Inverse of selection probability
* Design: Two-stage PPS sampling within strata
* Reference: Methodology Report Section 3.2
*============================================================
gen wgt_base = 1/prob_selection

*============================================================
* NON-RESPONSE ADJUSTMENT
* Propensity model estimated in 02_propensity.do
* Trimming at 3x median follows Potter (1990) recommendation
* to balance bias reduction against variance inflation
*============================================================
gen wgt_nr = wgt_base * (1/prop_response)
quietly summarize wgt_nr, detail
local trim_threshold = 3 * r(p50)
replace wgt_nr = min(wgt_nr, `trim_threshold')
```

The Comment Hierarchy

Three levels of documentation within code:

Level 1: File Header (REQUIRED for every script)

- Purpose, author, creation date, last modified
- Input files and output files
- Software dependencies and versions

Level 2: Section Headers (REQUIRED for each major block)

- What this section accomplishes
- Reference to methodology documentation

Level 3: Inline Comments (AS NEEDED)

- Explain non-obvious logic or edge cases
- Document "magic numbers" with sources

Template: File Header

```
/*============================================================
 SCRIPT:   03_calibration.do
 PROJECT:  ZQLFS 2024 Wave 2 Survey Weights

 PURPOSE:  Calibrate base weights to population benchmarks
           using iterative proportional fitting (raking)

 AUTHOR:   Dr. Lindiwe Moyo
 CREATED:  2024-05-15
 MODIFIED: 2024-06-10 (added province-level margins)

 INPUTS:   - 02_processed_data/zqlfs_basewgt.dta
           - 00_documentation/pop_benchmarks_2024.xlsx

 OUTPUTS:  - 03_weights/zqlfs_calwgt.dta

 REQUIRES: STATA 18 or later (uses ipfraking command)

 NOTES:    Population benchmarks from StatZambara mid-year
           estimates, released 2024-04-30
============================================================*/
```

Template: Section Headers

```
*============================================================
* SECTION 1: LOAD DATA AND VERIFY INPUTS
*============================================================
use "02_processed_data/zqlfs_basewgt.dta", clear
assert _N == 20940  // Expected household count

*============================================================
* SECTION 2: DEFINE CALIBRATION MARGINS
* Source: StatZambara mid-year population estimates 2024
* Reference: methodology_report.pdf, Section 4.3
*============================================================

* Province population totals (thousands)
matrix pop_province = (3245, 2876, 1543, 987, ///
                       2341, 1234, 876, 1098, 1567)
```

Never Leave "Magic Numbers" Unexplained

Bad practice:

```
replace wgt = wgt * 1.147
```

Good practice:

```
* Adjustment for sampling frame undercoverage
* Factor = Census population / Frame population
*        = 58,432,000 / 50,900,000 = 1.147
* Source: Frame Evaluation Report (2024), Table 2.3
replace wgt = wgt * 1.147
```

▶ Key Insight

Every number that isn't 0, 1, or 100 needs a documented source. Future analysts will thank you.

Document the "Why" Behind Decisions

Capture methodological choices and their rationale:

```
*============================================================
* METHODOLOGICAL DECISION: Weight Trimming Threshold
*------------------------------------------------------------
* Options considered:
*   (a) 3x median: Standard approach (Potter 1990, Ch. 4)
*   (b) 4x median: More conservative, less bias correction
*   (c) Fixed cap at 500: Used in previous ZQLFS wave
*
* Decision: 3x median
* Rationale: Given our attrition rate (17.3%), this provides
*            optimal bias-variance tradeoff. Affects only
*            0.7% of records (147 households).
* Approved by: Chief Methodologist, 2024-05-20
*============================================================
local trim_factor = 3
```

Version Control: Essential for Teams

Why use formal version control (e.g., Git)?

1. **Track all changes** — Who changed what, when, and why
2. **Revert mistakes** — Return to any previous working state
3. **Collaborate safely** — Multiple analysts, no file conflicts
4. **Branch experiments** — Test new methods without risk
5. **Audit trail** — Complete history for quality assurance

Minimum viable practice:

- Commit after each completed task
- Write meaningful commit messages
- Never commit broken code to the main branch

Writing Good Commit Messages

Bad commit messages:

```
"fixed stuff"
"updates"
"final version"
"asdfghjkl"
```

Good commit messages:

```
"Add province-level calibration margins"
"Fix trimming bug that excluded 147 records"
"Update benchmarks to 2024 mid-year estimates"
"Implement IPW for Wave 2 attrition adjustment"
```

▶ Key Insight

Pattern: Start with action verb + describe *what* changed and *why*.

Positive Example: GHS 2024 Data Quality

Zambara Case

Lab 5.3 generated an automated codebook for the GHS 2024 household file. The results demonstrate what excellent data management looks like.

GHS 2024 Household File Quality Metrics:

Metric	Value
Total variables	190
Total observations	20,940
Variables with 0% missing	**190 (100%)**
Variables with >5% missing	0
Variables with >25% missing	0

Documentation Quality Checklist

Before releasing survey weights, verify:

Naming:

- ☐ Variable names self-explanatory
- ☐ No version numbers in variables
- ☐ Consistent naming scheme used
- ☐ All weights follow standard pattern

Structure:

- ☐ Numbered folders/scripts
- ☐ README.md exists and is current
- ☐ Data flow documented

Comments:

- ☐ All scripts have file headers
- ☐ Section headers present
- ☐ All magic numbers sourced
- ☐ Key decisions documented

Validation:

- ☐ New analyst test passed
- ☐ Weights sum to benchmarks
- ☐ No extreme weights flagged

Module 5.3 Summary: Raising Your Bus Factor

The Three Pillars:

1. **NAMING:** Self-documenting variables
   Pattern: `type_level_stage_wave`
2. **STRUCTURE:** Numbered folders
   Always include README.md
3. **COMMENTS:** Explain WHY
   Headers + section markers + sources

The Ultimate Test:

Can a new analyst reproduce your weights in <1 hour without asking questions?

Quality Indicators:

- Every magic number sourced
- Decisions documented
- Data flow visualized

▶ Key Insight

Goal: Raise your organization's Bus Factor to ≥ 3.
Good documentation is not overhead—it is **professional obligation**.

BEST PRACTICE

Metadata Standards

Module 5.4: DDI and Public Documentation

Day 5: Synthesis and Best Practices

"Data without documentation is just noise."

Why Metadata Matters

Zambara Case

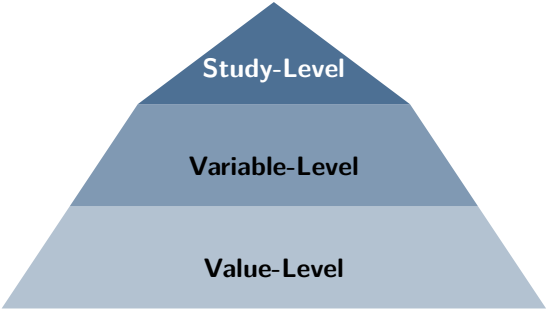
Lindiwe's team has produced excellent survey weights. Now the Ministry wants to release the microdata publicly. A researcher in Botswana downloads the file, but without documentation, they have no idea what `geotype=2` means or how to properly apply the weights.

Metadata transforms data into information:

- **Discoverability:** Users can find your survey in catalogs
- **Understanding:** Variables, codes, and concepts explained
- **Reproducibility:** Methodology transparent for replication
- **Interoperability:** Data can be combined across sources
- **Preservation:** Context survives beyond current staff

The Metadata Hierarchy

Three levels of survey metadata:



Title, abstract, methodology, geographic coverage, time period

Variable names, labels, types, universe, derivation

Code lists, category labels, missing value definitions

▶ Key Insight

Complete metadata addresses all three levels—from survey overview down to individual code values.

The DDI Standard

Data Documentation Initiative (DDI):

An international standard for describing social, behavioral, and economic data maintained by the DDI Alliance (www.ddialliance.org).

Why DDI matters for SADC:

- **International recognition:** Used by World Bank, IHSN, Eurostat, UN agencies
- **Software support:** NADA, Nesstar, Colectica, and other tools
- **Comparability:** Harmonized documentation across countries
- **Archival standard:** Ensures long-term preservation
- **SDMX compatibility:** Links to statistical data exchange

▶ Key Insight

DDI is the **lingua franca** of survey metadata—if you document in DDI, your data can be understood globally.

DDI Versions and Profiles

Two main DDI specifications:

DDI-Codebook (DDI-C)

- XML-based format
- Focus on data files
- Variable descriptions
- **Simpler to implement**
- Widely supported
- Best for: Single surveys

DDI-Lifecycle (DDI-L)

- Full lifecycle coverage
- Concept management
- Questionnaire flow
- Complex relationships
- Reusable components
- Best for: Survey programs

Recommendation: Start with DDI-Codebook for individual surveys. Move to DDI-Lifecycle when managing ongoing survey programs.

DDI-Codebook Structure

Key sections of a DDI-Codebook document:

Section	Content
`<docDscr>`	Document description (who created the metadata, when)
`<stdyDscr>`	Study description (title, abstract, methodology, scope)
`<fileDscr>`	File description (file names, formats, record counts)
`<dataDscr>`	Data description (variables, categories, statistics)
`<otherMat>`	Other materials (questionnaires, reports, manuals)

The `<dataDscr>` section contains `<var>` elements for each variable—this is where codebook detail lives.

DDI Variable Documentation

Essential elements for each variable:

```
<var ID="V25" name="geotype" intrvl="discrete">
  <labl>Geographic type of enumeration area</labl>
  <txt>Classification of the EA as urban, traditional,
       or farm based on census geography.</txt>
  <catgry>
    <catValu>1</catValu>
    <labl>Urban</labl>
    <catStat type="freq">12453</catStat>
  </catgry>
  <catgry>
    <catValu>2</catValu>
    <labl>Traditional</labl>
    <catStat type="freq">6892</catStat>
  </catgry>
  <catgry>
    <catValu>3</catValu>
    <labl>Farms</labl>
    <catStat type="freq">1595</catStat>
  </catgry>
  <varFormat type="numeric"/>
  <notes>Used for domain estimation and stratification.</notes>
</var>
```

Study-Level Metadata Elements

What users need to know about your survey:

Identification:

- Title and subtitle
- Unique identifier (DOI)
- Version number and date
- Producing agency

Scope:

- Geographic coverage
- Time period
- Universe/population
- Unit of analysis

Methodology:

- Sample design summary
- Data collection mode
- Response rates
- Weighting procedures

Access:

- Data access conditions
- Citation requirements
- Contact information
- Related publications

Documenting Survey Methodology

The methodology section must explain weighting clearly:

1. **Sampling design:** Stratification, stages, selection methods
2. **Sample size:** Planned vs. achieved, by domain
3. **Base weights:** How selection probabilities were calculated
4. **Adjustments applied:**
   - Non-response adjustment method
   - Calibration benchmarks and sources
   - Weight trimming (if applied)
5. **Design effects:** Key indicators and DEFF values
6. **Variance estimation:** Recommended method (Taylor/Replication)

△ Caution

Users who don't understand your weighting methodology will misuse your data. Be explicit and detailed.

Module 5.4: Learning Objectives

By the end of this module, you will be able to:

1. **Explain** the purpose and structure of DDI metadata standards
2. **Create** automated codebooks using R or STATA
3. **Document** variable-level metadata including codes and labels
4. **Produce** study-level documentation for public release
5. **Assess** data quality using automated diagnostics
6. **Design** user guides that enable proper data analysis

Lab 5.3 Connection: The automated codebook you generated demonstrates variable-level documentation in practice.

LAB 5.3 SHOWCASE

Automated Codebook Generation

GHS 2024 Household File Analysis

190 variables · 20,940 observations · 100% complete

Lab 5.3: Automated Codebook Results

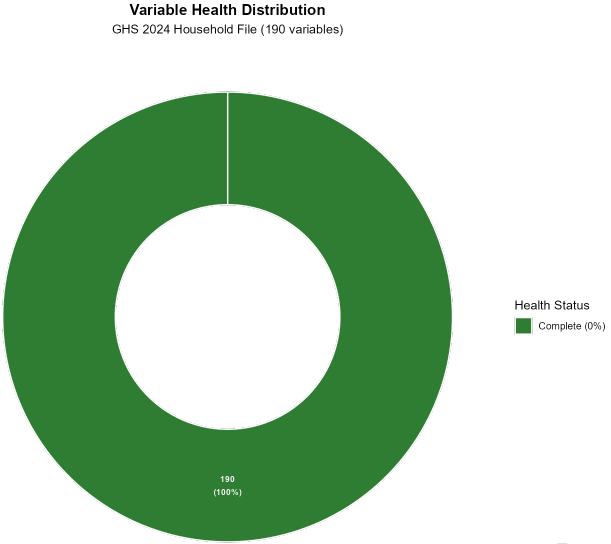
GHS 2024 Household File Summary:

Metric	Value
Dataset file	ghs-2024-hhold-v1.dta
Total observations	20,940 households
Total variables	190
Character variables	3
Numeric variables	187
Labelled variables	184
Overall completeness	**100.00%**
Variables with any missing	**0**

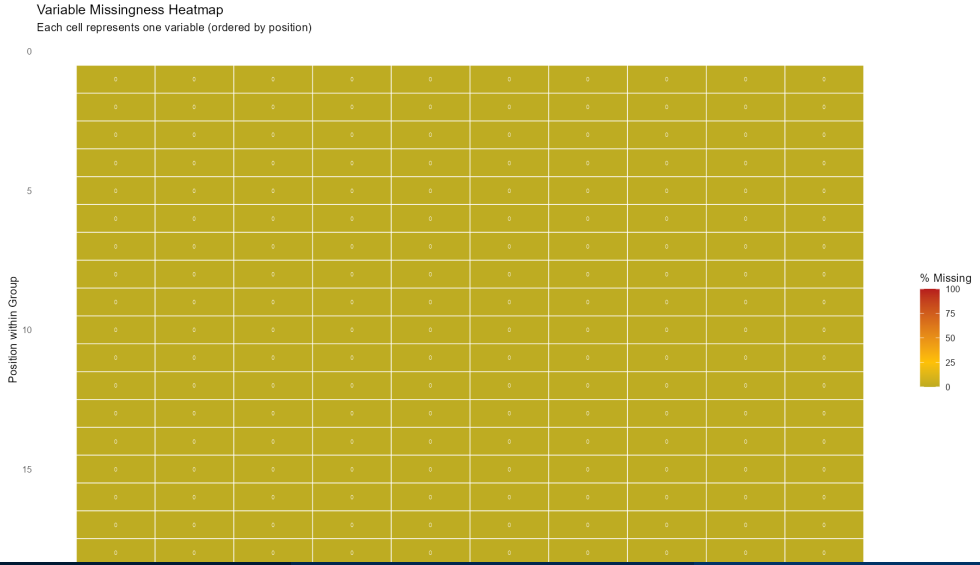
▶ Key Insight

Statistics South Africa achieves **perfect data completeness**—a gold standard demon-

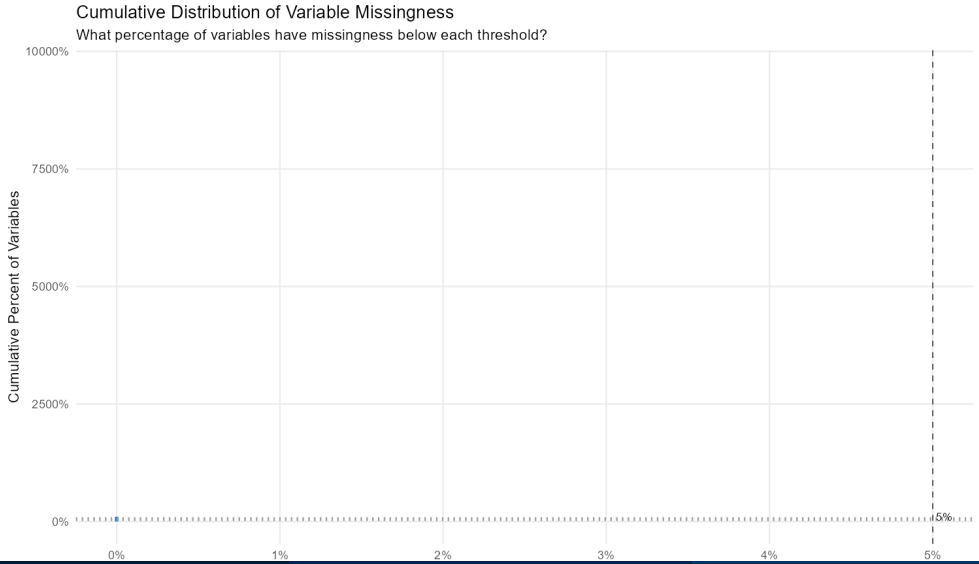
Data Quality Visualization: Health Distribution



Data Quality Visualization: Missingness Heatmap



Data Quality Visualization: Cumulative Distribution



Codebook Variable Inventory (Sample)

Excerpt from automated codebook (key variables of 190 total):

#	Variable	Label	Type	Unique
1	`uqnr`	Unique number	Character	20,940
3	`psu`	Primary sampling unit	Numeric	3,218
4	`prov`	Province	Labelled	9
6	`head_sex`	Sex of household head	Labelled	2
20	`wat_drinkwat`	Main source of water	Labelled	16
47	`eng_mainelect`	Main source of electricity	Labelled	7
105	`fsd_hungry`	Hunger	Labelled	3
170	`hholdsz`	Household size	Labelled	21
185	`stratum`	Stratum	Character	248
186	`geotype`	Geographical type	Labelled	3
190	`house_wgt`	House Weight	Numeric	4,292

Key sampling variables (`psu`, `stratum`, `house_wgt`) are clearly documented for proper analysis.

From Codebook to User Guide

A codebook documents *what* is in the data.
A user guide explains *how* to use the data.

Codebook contains:

- Variable names and labels
- Code values and categories
- Missing value definitions
- Basic statistics

User guide adds:

- How to apply weights
- Recommended software setup
- Common analysis examples
- Known limitations
- Citation requirements

▶ Key Insight

Public data releases need **both**: technical codebook + practical user guide.

User Guide: Recommended Structure

Essential sections for a public microdata user guide:

1. **Introduction** — Survey purpose, population, time period
2. **Sample Design** — Stratification, stages, selection overview
3. **Data Collection** — Mode, dates, response rates
4. **File Structure** — Available files, record layouts, linking keys
5. **Using the Weights** — Which weight for which analysis
6. **Variance Estimation** — Required variables and methods
7. **Data Quality Notes** — Known issues, limitations, cautions
8. **Getting Started** — Code examples in R and STATA
9. **Citation and Terms** — Required citation, access conditions
10. **Contact Information** — Technical support resources

User Guide: The Weights Section

This section prevents the most common user errors:

```
## 5. USING THE SURVEY WEIGHTS

### Available Weight Variables
- `house_wgt`: Household-level weight for household analyses
- `person_wgt`: Person-level weight for individual analyses

### When to Use Which Weight
| Analysis Type              | Weight Variable | Example                |
|---------------------------|-----------------|------------------------|
| Household characteristics | house_wgt       | Dwelling type, income  |
| Individual characteristics| person_wgt      | Employment, education  |
| Person count from HH file | house_wgt       | Household size totals  |

### IMPORTANT: Never analyze without weights!
Unweighted analyses will produce BIASED estimates that do not
represent the national population. Always use svy commands.

### Software Setup
R:     svydesign(ids=~psu, strata=~stratum, weights=~house_wgt)
STATA: svyset psu [pw=house_wgt], strata(stratum)
```

User Guide: Variance Estimation Section

Tell users exactly how to compute correct standard errors:

```
## 6. VARIANCE ESTIMATION

### Required Variables for Correct Standard Errors
- `psu`: Primary Sampling Unit identifier (cluster variable)
- `stratum`: Sampling stratum identifier
- `house_wgt` or `person_wgt`: Analysis weight

### Recommended Method
Taylor Series Linearization is the default in most software.
This survey has 3,218 PSUs in 248 strata.

### Singleton PSU Handling
Some strata contain only one sampled PSU. Software options:
- R (survey): options(survey.lonely.psu = "adjust")
- STATA: svyset ..., singleunit(certainty)

### WARNING
Standard errors computed WITHOUT accounting for clustering
will be UNDERESTIMATED, typically by 10-20%. This leads to
false precision and invalid confidence intervals.
```

User Guide: Getting Started Code

Provide copy-paste examples that work immediately:

R Example:

```r
library(survey)
library(haven)

# Load data
ghs <- read_dta("ghs2024_hh.dta")

# Define survey design
ghs_des <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~house_wgt,
  data = ghs
)

# Estimate mean HH size
svymean(~hholdsz, ghs_des)
```

STATA Example:

```stata
* Load data
use "ghs2024_hh.dta", clear

* Define survey design
svyset psu [pw=house_wgt], ///
  strata(stratum)

* Estimate mean HH size
svy: mean hholdsz

* Tabulation by province
svy: tab prov, count
```

User Guide: Data Quality Notes

Be transparent about limitations—users will trust you more:

1. **Coverage limitations**
   - Institutional population excluded (hostels, prisons, etc.)
   - New settlements since 2022 census may be under-covered
2. **Non-response patterns**
   - Overall response rate: 92.3%
   - Higher refusal in urban high-income areas
   - Weights adjusted for differential non-response
3. **Item-specific issues**
   - Income variables subject to recall error
   - Some variables applicable only to subsets (universe restrictions)
4. **Confidentiality edits**
   - Top-coding applied to income above R500,000/month
   - Some small-area geography suppressed

Tools for Metadata Creation

Software for creating DDI-compliant documentation:

Tool	Cost	Best For
Nesstar Publisher	Free	Full DDI-C authoring
IHSN Metadata Editor	Free	World Bank/IHSN standards
Colectica	Commercial	DDI-L lifecycle management
R (labelled, sjlabelled)	Free	Automated extraction
STATA (codebook)	Licensed	Quick variable summaries
Python (pandas_profiling)	Free	Automated reports

Recommended workflow:

1. Extract variable metadata automatically (R/STATA)
2. Add study-level documentation manually
3. Export to DDI XML for archiving
4. Generate PDF user guide from same source

NADA: National Data Archive Platform

NADA is the standard platform for survey data dissemination:

- Developed by IHSN (International Household Survey Network)
- Used by 100+ countries and organizations worldwide
- Imports DDI metadata directly
- Provides searchable catalog interface
- Manages data access requests
- Tracks citations and usage

SADC members using NADA:
South Africa (DataFirst), Botswana, Malawi, Zambia, Zimbabwe, Tanzania, Mozambique

▶ Key Insight

Publishing to NADA makes your survey discoverable to researchers globally through the IHSN catalog.

Metadata Quality Checklist

Before public release, verify documentation completeness:

Study-Level:

- ☐ Title and abstract complete
- ☐ Geographic scope defined
- ☐ Time period specified
- ☐ Sample design described
- ☐ Weighting documented
- ☐ Response rates reported

Variable-Level:

- ☐ All variables labeled
- ☐ All codes explained
- ☐ Missing values defined
- ☐ Universe noted where applicable
- ☐ Derived variables explained

User Guide:

- ☐ Weight usage instructions clear
- ☐ Variance estimation explained
- ☐ Working code examples provided
- ☐ Citation requirements stated

Common Metadata Failures

Mistakes that frustrate data users:

1. **Missing weight documentation**
   - × "Use house_wgt for analysis" (which analysis?)
   - ✓ Table showing which weight for which file/level
2. **Undefined codes**
   - × Values 8, 9, 99 appear but aren't explained
   - ✓ Explicit missing value definitions (8=N/A, 9=Refused)
3. **No variance estimation guidance**
   - × PSU and stratum variables exist but aren't mentioned
   - ✓ Explicit svyset/svydesign code provided
4. **Outdated documentation**
   - × Codebook from v1.0, data file is v2.3
   - ✓ Version tracking and change logs

Zambara ZQLFS: User Guide Outline

Zambara Case

Lindiwe's team prepares the ZQLFS public release. Here is their user guide table of contents:

1. Introduction
2. Background and Objectives
3. Sample Design
   - 3.1 Stratification
   - 3.2 Sample Size
   - 3.3 Selection Procedures
4. Data Collection
5. Data Processing
6. Weighting Methodology
   - 6.1 Base Weights
   - 6.2 Non-Response Adjustment
   - 6.3 Calibration
7. Using the Data Files
8. Variance Estimation
9. Data Quality
10. References and Citation

Exercise 5.4: Create a Mini User Guide

Task: Using the GHS 2024 codebook from Lab 5.3, draft a 2-page user guide section on "Using the Survey Weights."

Your section must include:

1. List of weight variables with descriptions
2. Table: which weight for which analysis type
3. Required variables for variance estimation
4. Complete code example (R or STATA) that:
   - Loads the data
   - Sets up the survey design
   - Estimates mean household size with confidence interval
   - Produces a weighted frequency table by province
5. One paragraph on what happens if weights are ignored

Time: 20 minutes | **Deliverable:** Markdown or Word document

Module 5.4 Summary: Metadata Standards

Key Concepts:

- DDI is the international standard for survey documentation
- Three levels: study, variable, value
- Codebook $\neq$ User Guide (need both)
- Automated tools can extract variable metadata
- NADA enables global discoverability

User Guide Essentials:

- Weight selection guidance
- Variance estimation setup
- Working code examples
- Known limitations
- Citation requirements

▶ Key Insight

The test of good metadata: Can a researcher in another country correctly analyze your data using only the documentation provided?

Looking Ahead: Workshop Synthesis

We've covered the complete survey lifecycle:

Final Module (5.5): Workshop synthesis, implementation roadmap, and next steps for your NSO.

WORKSHOP SYNTHESIS

The Implementation Roadmap

Module 5.5: From Learning to Action

Day 5: Synthesis and Best Practices

"Knowledge without action is merely entertainment."

The Post-Workshop Challenge

△ Caution

The Workshop Graveyard:

Studies show that 70% of training content is forgotten within 24 hours, and 90% within a week—unless deliberately applied. Most workshop participants return to their offices with good intentions but face immediate pressures that crowd out implementation.

Common barriers to implementation:

- Urgent deadlines waiting at the office
- No dedicated time allocated for methodology improvement
- Lack of management buy-in for changes
- "We've always done it this way" resistance
- Technical skills learned but not practiced

The Solution: A Structured Roadmap

Transform workshop learning into organizational change:

AUDIT	PILOT	CALIBRATE	LAUNCH
Month 1-2	Month 3-4	Month 5-6	Month 7+
Assess current practice	Test new methods	Refine and document	Full production

▶ Key Insight

A 6-month implementation timeline transforms workshop skills into institutional capability.

Lindiwe's Implementation Journey

Zambara Case

The workshop concludes. Lindiwe returns to ZNSO with a notebook full of techniques and a laptop full of code. Her Director asks: "What will you actually change?"

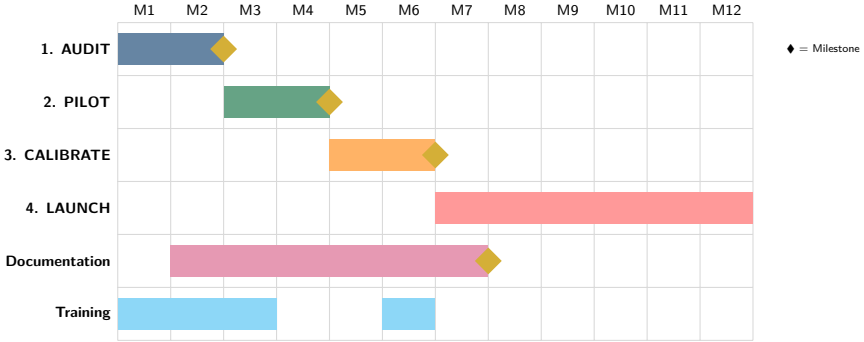
Lindiwe presents a 6-month roadmap. By month 7, Zambara's Quarterly Labour Force Survey launches with proper stratification, calibrated weights, documented methodology, and—most importantly—a team trained to maintain it.

Your goal: Create your own implementation roadmap before leaving this workshop.

This module provides:

- Phase-by-phase implementation guide
- Checklists for each milestone
- Templates for management buy-in
- Risk mitigation strategies

Implementation Gantt Chart: 6-Month Overview



Four phases with parallel documentation and training tracks. Gold diamonds mark key deliverables.

PHASE 1

AUDIT
Current Practice

Months 1–2

"You can't improve what you don't understand."

Phase 1: Audit Objectives

Before changing anything, understand what you have:

1. **Frame Assessment**
   - When was the sampling frame last updated?
   - What is the estimated coverage rate?
   - Are there known gaps (new settlements, displaced populations)?
2. **Methodology Review**
   - Document current stratification approach
   - Review existing weighting procedures
   - Identify variance estimation practices (or lack thereof)
3. **Capacity Assessment**
   - Who has R/STATA survey analysis skills?
   - What documentation exists for current methods?
   - What is the Bus Factor for key processes?

Audit Checklist: Sampling Frame

Evaluate your sampling frame against best practices:

Coverage:

- ☐ Frame source identified (census, register)
- ☐ Reference date documented
- ☐ Geographic completeness assessed
- ☐ Population groups covered/excluded
- ☐ Under-coverage estimate available

Quality:

- ☐ Duplicate PSUs identified
- ☐ Size measures current and accurate
- ☐ Stratification variables available
- ☐ Contact information usable
- ☐ Update procedures documented

Score your frame:

- 8–10 checks: Ready for advanced methods
- 5–7 checks: Frame improvement needed first
- <5 checks: Major frame development required

Audit Checklist: Current Methodology

Document your current survey methodology:

Component	Documented?	Follows Best Practice?
Sample design (strata, stages)	☐ Yes / ☐ No	☐ Yes / ☐ Partially / ☐ No
Selection probabilities	☐ Yes / ☐ No	☐ Yes / ☐ Partially / ☐ No
Base weight calculation	☐ Yes / ☐ No	☐ Yes / ☐ Partially / ☐ No
Non-response adjustment	☐ Yes / ☐ No	☐ Yes / ☐ Partially / ☐ No
Calibration method	☐ Yes / ☐ No	☐ Yes / ☐ Partially / ☐ No
Variance estimation	☐ Yes / ☐ No	☐ Yes / ☐ Partially / ☐ No
Weight validation	☐ Yes / ☐ No	☐ Yes / ☐ Partially / ☐ No

Priority rule: Fix documentation gaps before methodology gaps.

Audit Deliverable: Gap Analysis

Create a gap analysis document:

Area	Current State	Target State	Priority
Sampling frame	2011 census PSUs	Census 2022 + satellite update	High
Stratification	Province only	Province × urban/rural	Medium
Weighting	Base weights only	Full calibration	High
Variance estimation	Not done	Taylor linearization	High
Documentation	Minimal	Full DDI codebook	Medium

▶ Key Insight

The gap analysis becomes your implementation priority list. Present it to management for resource allocation.

Phase 1 Milestone: Audit Report

End of Month 2 deliverable:

Audit Report Contents:

1. Executive summary (1 page for management)
2. Frame assessment with coverage estimates
3. Current methodology documentation
4. Gap analysis with priorities
5. Resource requirements for improvements
6. Proposed pilot design (preview of Phase 2)

Success criteria:

- ✓ Management has approved pilot phase
- ✓ Resources allocated (staff time, software)
- ✓ Pilot survey identified
- ✓ Implementation team assigned

PHASE 2

PILOT
New Methods

Months 3–4

"Test in small scale before betting the organization."

Phase 2: Pilot Design Principles

Select a pilot that minimizes risk while maximizing learning:

Good pilot candidates:

- Smaller survey with shorter timeline
- Survey with known issues to fix
- Survey scheduled within the pilot window
- Survey where failure is recoverable

Avoid piloting with:

- Flagship surveys with political visibility
- Surveys feeding international indicators (SDGs)
- Surveys with no margin for timeline slippage

▶ Key Insight

The pilot should be consequential enough to test real methods, but not so critical that

Pilot Implementation Steps

Month 3: Design and Preparation

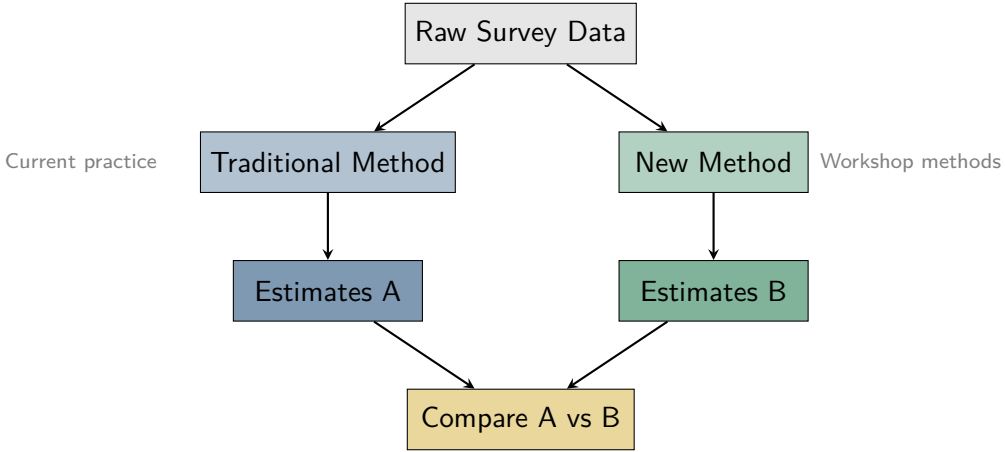
1. Apply workshop methods to pilot survey design
   - Implement improved stratification
   - Calculate proper selection probabilities
   - Set up survey design in R/STATA
2. Create parallel processing tracks
   - Traditional method (backup)
   - New method (pilot)
3. Document methodology decisions

Month 4: Execution and Comparison

1. Process pilot survey with new methods
2. Compare results: new vs. traditional
3. Document deviations and decisions
4. Prepare pilot evaluation report

Risk Mitigation: Parallel Processing

Run old and new methods simultaneously:



Evaluating Pilot Results

Compare new vs. traditional methods on:

Criterion	Traditional	New Method
Estimates align with benchmarks	☐	☐
Standard errors reasonable	☐	☐
Weights sum to population	☐	☐
No extreme weights	☐	☐
Design effects as expected	☐	☐
Results are explainable	☐	☐
Process is documented	☐	☐
Staff can reproduce	☐	☐

▶ Key Insight

The new method should perform **at least as well** as traditional—with better documen-

Phase 2 Milestone: Pilot Evaluation Report

End of Month 4 deliverable:

Pilot Evaluation Report Contents:

1. Methodology comparison summary
2. Side-by-side estimate comparisons
3. Variance estimation validation
4. Process documentation assessment
5. Lessons learned and adjustments needed
6. Recommendation: proceed / revise / abandon

Decision gate:

- **Green:** Proceed to calibration phase
- **Yellow:** Revise and re-pilot
- **Red:** Return to audit phase

PHASE 3

CALIBRATE and Refine

Months 5–6

"Fine-tune the engine before the race."

Phase 3: Calibration Activities

Refine methods based on pilot lessons:

Month 5: Technical Refinement

- Incorporate pilot feedback into methodology
- Finalize calibration benchmarks and sources
- Optimize weight trimming thresholds
- Validate variance estimation procedures
- Test edge cases and error handling

Month 6: Documentation and Training

- Complete technical methodology report
- Create user guide and codebook templates
- Train additional staff on new methods
- Develop standard operating procedures (SOPs)
- Prepare production scripts and templates

Calibration: Finalizing Benchmarks

Secure official population benchmarks:

Benchmark	Source	Verified?
Total population	Mid-year estimates 2024	☐
Province totals	Mid-year estimates 2024	☐
Urban/rural split	Census 2022 projections	☐
Age-sex distribution	Census 2022 projections	☐
Household totals	Census 2022 + growth rate	☐

Critical actions:

- Obtain official written authorization for benchmark use
- Document benchmark vintage and any adjustments
- Establish process for benchmark updates

Standard Operating Procedures (SOPs)

Convert ad-hoc processes into documented procedures:

Essential SOPs to develop:

1. **Sample Selection SOP**
   - Step-by-step selection process
   - Script templates with comments
   - Quality checks at each stage
2. **Weighting SOP**
   - Base weight calculation
   - Non-response adjustment triggers
   - Calibration procedure and validation
3. **Documentation SOP**
   - Required metadata elements
   - Codebook generation workflow
   - User guide template

Phase 3 Milestone: Production Readiness

End of Month 6 deliverable:

Production Readiness Package:

1. Final methodology documentation
2. Validated R/STATA scripts with comments
3. Standard Operating Procedures manual
4. User guide and codebook templates
5. Training materials for ongoing use
6. Quality assurance checklists

Readiness criteria:

- ✓ At least 2 staff can execute full process independently
- ✓ All scripts tested on current and historical data
- ✓ Management sign-off on methodology
- ✓ Calendar scheduled for first production run

PHASE 4

LAUNCH Production

Month 7 and Beyond

"Now the real work begins."

Phase 4: Launch Preparation

Final checks before production:

Technical Readiness:

- ☐ Scripts version-controlled
- ☐ Input data pathways confirmed
- ☐ Output file formats specified
- ☐ Error handling tested
- ☐ Backup procedures in place

Organizational Readiness:

- ☐ Roles and responsibilities clear
- ☐ Timeline communicated
- ☐ Stakeholders informed
- ☐ Support plan for issues
- ☐ Success metrics defined

▶ Key Insight

Launch is not an event—it's the beginning of continuous improvement.

First Production Run: What to Expect

The first production run will reveal issues. Plan for them:

Common first-run challenges:

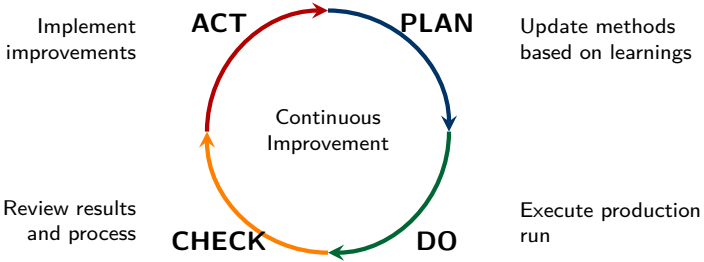
- Data format differences from pilot
- Edge cases not encountered in testing
- Integration issues with existing workflows
- Timeline pressure creating shortcuts

Mitigation strategies:

1. Schedule extra time for first production (add 50% buffer)
2. Keep pilot lead available for troubleshooting
3. Document all issues for future reference
4. Conduct immediate post-mortem review
5. Update SOPs with lessons learned

Beyond Launch: Continuous Improvement

Build improvement into regular operations:



After each survey round, review what worked and what didn't. Update SOPs accordingly.

Measuring Implementation Success

Track these metrics to demonstrate value:

Metric	Baseline	Target (Year 1)
Design effect documentation	None	All surveys
Staff with R/STATA skills	1–2	4–5
Surveys with full codebooks	20%	80%
Weight validation checks	Ad-hoc	Standardized
Variance estimates reported	Rarely	Always
Bus Factor	1	3+

▶ Key Insight

Quantified improvements help justify continued investment and demonstrate workshop ROI.

Implementation Risk Register

Anticipate and plan for common risks:

Risk	Likelihood	Mitigation
Key staff leave	Medium	Cross-train multiple staff; document everything
Management support wanes	Medium	Regular progress reports; quantify benefits
Software/license issues	Low	Plan for open-source alternatives (R)
Data format changes	Medium	Build flexible scripts; validate inputs
Timeline slippage	High	Build buffer into schedule; phase deliverables
Resistance to change	Medium	Involve staff early; celebrate wins

Update risk register monthly during implementation.

Exercise 5.5: Your Implementation Roadmap

Create your personal 6-month implementation plan:

Complete these elements (20 minutes):

1. **Audit priorities** (list top 3 gaps in your NSO)
2. **Pilot candidate** (which survey will you test with?)
3. **Resources needed** (staff, software, time)
4. **Key stakeholders** (who needs to approve?)
5. **First milestone** (what will you deliver by Month 2?)
6. **Success metric** (how will you measure improvement?)

Share with your workshop neighbor: Get feedback on feasibility and blind spots.

Deliverable: One-page implementation summary to take home.

Module 5.5 Summary: From Learning to Action

The Four Phases:

1. **AUDIT** (M1–2): Assess current state
2. **PILOT** (M3–4): Test new methods
3. **CALIBRATE** (M5–6): Refine and document
4. **LAUNCH** (M7+): Production and improve

Critical Success Factors:

- Management buy-in from Day 1
- Parallel processing during pilot
- Documentation at every stage
- Multiple trained staff (Bus Factor ≥ 3)

▶ Key Insight

Your homework:

Within 2 weeks of returning:
☐ Complete audit checklist
☐ Identify pilot survey
☐ Brief your director
☐ Schedule first team meeting

"The best time to implement was yesterday. The second best time is today."

WORKSHOP FINALE

The Golden Rules

Module 5.6: Rapid-Fire Review

Day 5: Synthesis and Best Practices

"Everything you need to remember, nothing you can afford to forget."

The Ministerial Briefing

Zambara Case

Six months have passed. Lindiwe stands in the Minister's boardroom, laptop connected to the projector. Around the table sit the Permanent Secretary, the Director of Statistics, and representatives from the World Bank and SADC Secretariat.

"Minister," Lindiwe begins, "I am pleased to present the **Zambara Survey Design Protocol**—a comprehensive methodology framework that will ensure our household surveys meet international standards for the next decade."

She clicks to the first slide: **The Golden Rules of Survey Methodology**.

What follows: A rapid-fire review of the essential concepts from this workshop.

GOLDEN RULES FROM

DAY 1

Sample Design Foundations

Design Effects · Stratification · Allocation · Frames

Golden Rule #1: Why Not Simple Random Sampling?

Question

Why do we use complex multi-stage designs instead of simple random sampling for household surveys?

Golden Rule #1: Why Not Simple Random Sampling?

Question

Why do we use complex multi-stage designs instead of simple random sampling for household surveys?

Golden Rule

Cost and feasibility.

SRS would require a complete list of all households (rarely available) and travel to randomly scattered locations (prohibitively expensive).

Clustering reduces costs; stratification improves precision.

Golden Rule #2: What Is the Design Effect?

Question

Define the design effect (DEFF) and explain what DEFF $= 2.5$ means.

Golden Rule #2: What Is the Design Effect?

Question

Define the design effect (DEFF) and explain what DEFF = 2.5 means.

Golden Rule

DEFF = Var(complex) / Var(SRS)

DEFF = 2.5 means you need **2.5 times as many** observations to achieve the same precision as SRS.

Effective sample size = n / DEFF

Golden Rule #3: What Drives the Design Effect?

Question

Write the formula for DEFF due to clustering and identify which factor you can control.

Golden Rule #3: What Drives the Design Effect?

Question

Write the formula for DEFF due to clustering and identify which factor you can control.

Golden Rule

$$\mathbf{DEFF} \approx \mathbf{1 + (b - 1)}\rho$$

Where:

b = cluster size (controllable)

ρ = intra-cluster correlation (fixed)

Smaller clusters = smaller DEFF

Golden Rule #4: Why Stratify?

Question

What are the three main benefits of stratification in sample design?

Golden Rule #4: Why Stratify?

Question

What are the three main benefits of stratification in sample design?

Golden Rule

1. **Precision:** Reduce variance by grouping similar units
2. **Domains:** Guarantee minimum sample in each subgroup
3. **Admin:** Organize fieldwork by region

Stratify by variables correlated with key outcomes

Golden Rule #5: Frame Errors

Question

Name the four types of sampling frame errors.

Golden Rule #5: Frame Errors

Question

Name the four types of sampling frame errors.

Golden Rule

Under-coverage: Units missing from frame
Over-coverage: Non-existent units listed
Duplication: Same unit listed multiple times
Ineligibles: Out-of-scope units included

Frame errors cause coverage bias—weights cannot fully fix

GOLDEN RULES FROM

DAY 2

Computational Tools

R/STATA Setup · GIS Integration · Quality Checks

Golden Rule #6: Always Declare the Design

Question

What three elements must you specify when declaring a survey design in R or STATA?

Golden Rule #6: Always Declare the Design

Question

What three elements must you specify when declaring a survey design in R or STATA?

Golden Rule

1. **Cluster ID** (PSU variable)
2. **Strata** (stratification variable)
3. **Weights** (analysis weight)

```
svydesign(ids=~psu,
strata=~stratum, weights=~wgt)
```

Never analyze without declaring design

Golden Rule #7: The Cost of Ignoring Design

Question

What happens to standard errors if you ignore the complex survey design?

Golden Rule #7: The Cost of Ignoring Design

Question

What happens to standard errors if you ignore the complex survey design?

Golden Rule

Standard errors are UNDERESTIMATED

Typically by 10–30% (we found 13.6% in GHS 2024)

Consequence: False precision, invalid confidence intervals, wrong conclusions

Golden Rule #8: Handle Singleton PSUs

Question

What is a singleton PSU and how should you handle it?

Golden Rule #8: Handle Singleton PSUs

Question

What is a singleton PSU and how should you handle it?

Golden Rule

Singleton: Only one PSU sampled in a stratum

Solutions:

R: `options(survey.lonely.psu="adjust'`

STATA: `singleunit(certainty)`

Never let singletons crash variance estimation

GOLDEN RULES FROM

DAY 3

Weighting and Variance

Base Weights · Calibration · Taylor Series · Replication

Golden Rule #9: Base Weight = Inverse Probability

Question

What is a base weight and why is it calculated as the inverse of selection probability?

Golden Rule #9: Base Weight = Inverse Probability

Question

What is a base weight and why is it calculated as the inverse of selection probability?

Golden Rule

Base weight = 1 / π_i

Each sampled unit "represents" $1/\pi_i$ units in the population.

If selection probability = 1/500, that unit represents 500 population units

Golden Rule #10: The Weight Equation

Question

What are the three main components of a final survey weight?

Golden Rule #10: The Weight Equation

Question

What are the three main components of a final survey weight?

Golden Rule

Final Weight =

Base Weight
$\times$ Non-Response Adjustment
$\times$ Calibration Factor

Each component corrects a different source of bias

Golden Rule #11: Why Calibrate?

Question

What does calibration accomplish and what external data does it require?

Golden Rule #11: Why Calibrate?

Question

What does calibration accomplish and what external data does it require?

Golden Rule

Calibration: Adjusts weights so weighted totals match known population totals

Requires: Reliable benchmarks (census, admin data, projections)

Reduces bias AND often reduces variance

Golden Rule #12: Trim Extreme Weights

Question

Why trim extreme weights and what is a common threshold?

Golden Rule #12: Trim Extreme Weights

Question

Why trim extreme weights and what is a common threshold?

Golden Rule

Why: Extreme weights inflate variance

Threshold: 3–5$\times$ median weight

Small bias introduced, large variance reduction—usually good tradeoff

Golden Rule #13: Taylor vs. Replication

Question

When use Taylor linearization vs. replication methods?

Golden Rule #13: Taylor vs. Replication

Question

When use Taylor linearization vs. replication methods?

Golden Rule

Taylor: Fast; analytical; needs design vars

Replication: Flexible; works for any statistic; computationally intensive

Taylor for routine; Replication for complex estimators

GOLDEN RULES FROM

DAY 4

Error Mitigation

TSE Framework · Non-Response · Imputation

Golden Rule #14: Total Survey Error

Question

Name the two main branches of Total Survey Error.

Golden Rule #14: Total Survey Error

Question

Name the two main branches of Total Survey Error.

Golden Rule

Representation errors: Coverage, sampling, non-response, adjustment

Measurement errors: Validity, questionnaire, interviewer, respondent, processing

Optimize total error, not just one component

Golden Rule #15: Non-Response Creates Bias When...

Question

When does unit non-response cause bias?

Golden Rule #15: Non-Response Creates Bias When...

Question

When does unit non-response cause bias?

Golden Rule

Bias occurs when:
Non-respondents differ systematically from respondents on the variable of interest

High response rate $\neq$ No bias

It's the correlation that matters

Golden Rule #16: Imputation Best Practices

Question

What are key principles for valid imputation?

Golden Rule #16: Imputation Best Practices

Question

What are key principles for valid imputation?

Golden Rule

1. Preserve distributions, not just means
2. Maintain variable relationships
3. Incorporate randomness
4. Flag imputed values

Hot-deck and regression are workhorses

GOLDEN RULES FROM

DAY 5

Longitudinal & Documentation

Panel Design · Attrition · Bus Factor · Metadata

Golden Rule #17: Attrition Is Not Random

Question

How do you correct for attrition bias in longitudinal surveys?

Golden Rule #17: Attrition Is Not Random

Question

How do you correct for attrition bias in longitudinal surveys?

Golden Rule

Inverse Probability Weighting:

$w_{long} = w_{base} \times (1/\hat{p}_{stay})$

Upweight survivors who "look like" dropouts

Urban renters often have highest attrition

Golden Rule #18: Raise Your Bus Factor

Question

What is the "Bus Factor" and what is the minimum acceptable level?

Golden Rule #18: Raise Your Bus Factor

Question

What is the "Bus Factor" and what is the minimum acceptable level?

Golden Rule

Bus Factor: Number of people who could leave before critical knowledge is lost

Minimum: 3

Fix with: documentation, training, version control

Golden Rule #19: Document in DDI

Question

What is DDI and why use it?

Golden Rule #19: Document in DDI

Question

What is DDI and why use it?

Golden Rule

DDI: Data Documentation Initiative—international metadata standard

Global discoverability, software interoperability, long-term preservation

DDI = lingua franca of survey metadata

The Minister's Response

Zambara Case

Lindiwe concludes her presentation. The room is silent for a moment.

The Minister nods slowly. "Dr. Moyo, this is exactly what Zambara needs. You've shown us not just what to do, but *why* it matters. The World Bank representative has already indicated this protocol could serve as a model for the region."

The Permanent Secretary adds: "What impressed me most is the documentation. For the first time, our methodology will survive staff turnover."

"When can you begin implementation?" the Minister asks.

Lindiwe smiles. "Minister, we already have. The ZQLFS pilot launched last month. Full production begins in Q3."

The Zambara Survey Design Protocol

Design Principles:

- Multi-stage stratified sampling
- Frame updated with GIS/satellite
- Optimal allocation for domains
- Design effects monitored

Weighting Protocol:

- Base weights from selection prob
- Propensity-based NR adjustment
- GREG calibration to benchmarks
- Trimming at $3\times$ median

Quality Assurance:

- TSE framework for monitoring
- Variance estimation mandatory
- Imputation with flags

Documentation:

- DDI-compliant codebooks
- Full user guides
- SOPs for all processes
- Bus Factor ≥ 3

Your Final Challenge

▶ Key Insight

The Golden Rules Memory Test

Write down from memory:

1. The DEFF formula
2. The three weight components
3. The three elements of survey design declaration
4. One cause of non-response bias
5. Your organization's current Bus Factor

If you recall these in six months, the workshop succeeded.

But more importantly: **Will you implement what you've learned?**

"The goal of survey methodology is not mathematical elegance, but truth—the truth about how people live, what they need, and how policy can improve their lives."

— Adapted from Leslie Kish

Thank you for your commitment to excellence in survey methodology.

★ CLOSING CEREMONY ★

Congratulations, Survey Methodologists!

SADC Regional Training Workshop

Advanced Sampling Methods

March 2026 · Johannesburg, South Africa

What We've Accomplished Together

Five days of intensive learning:

300 slides	**15+** lab exercises	**19** golden rules	**1** Lindiwe success story

Skills gained:

- Design complex stratified multi-stage samples
- Implement proper weighting and calibration
- Estimate variance correctly using R and STATA
- Document methodology to international standards
- Build longitudinal surveys with attrition correction

THE ULTIMATE TEST

Final Challenge
5 Hard Questions

No notes. No neighbors. Just you and your knowledge.

"Pressure makes diamonds."

Get 4/5 correct to earn your certificate with distinction!

Final Quiz: Question 1 of 5

CALCULATION CHALLENGE

A survey has 200 PSUs with an average of 15 households per PSU. The intra-cluster correlation (ρ) for household income is 0.08.

Calculate the design effect (DEFF) and effective sample size.

Write your answers:

- DEFF = _____
- Effective sample size = _____

Time: 90 seconds

Final Quiz: Answer 1

SOLUTION

DEFF $= 1 + (b - 1)\rho = 1 + (15 - 1) \times 0.08 = 1 + 1.12 =$ **2.12**

Total sample $= 200$ PSUs $\times$ 15 HH $= 3{,}000$ households

Effective sample size $=$ n / DEFF $= 3{,}000 / 2.12 =$ **1,415 households**

▶ Key Insight

Interpretation: The clustering means you have the precision equivalent of only 1,415 households under SRS, even though you surveyed 3,000.

Did you get it? ✓ Yes / × No

Final Quiz: Question 2 of 5

CONCEPTUAL CHALLENGE

A national survey achieves a 95% response rate. The survey director claims: "With such high response, non-response bias is negligible."

Is the director correct? Explain why or why not in one sentence.

Your answer:

_____

_____

Time: 60 seconds

Final Quiz: Answer 2

SOLUTION

The director is INCORRECT.

Non-response bias depends on the **correlation between response propensity and the outcome variable**, not just the response rate.

Even with 95% response, if the 5% who refused systematically differ on key variables (e.g., all high-income or all unemployed), bias can be substantial.

▶ Key Insight

Remember: High response rate $\neq$ Low bias.
A 70% response rate with random missingness may have less bias than 95% with systematic refusal patterns.

Final Quiz: Question 3 of 5

CODE CHALLENGE

Write the R code to properly declare the GHS 2024 survey design using:

- PSU variable: `psu`
- Stratification: `stratum`
- Weight: `house_wgt`
- Data frame: `ghs_hh`

Include the option to handle singleton PSUs.

Your code:

Time: 90 seconds

Final Quiz: Answer 3

SOLUTION

```
# Handle singleton PSUs
options(survey.lonely.psu = "adjust")

# Declare survey design
ghs_design <- svydesign(
  ids = ~psu,
  strata = ~stratum,
  weights = ~house_wgt,
  data = ghs_hh
)
```

Key elements:

- `ids = ~psu` — Cluster variable

Final Quiz: Question 4 of 5

WEIGHTING CHALLENGE

Put these weight components in the correct ORDER of application:

A. Calibration adjustment
B. Base weight (inverse selection probability)
C. Weight trimming
D. Non-response adjustment

Correct order:

_____ → _____ → _____ → _____

Time: 45 seconds

Final Quiz: Answer 4

SOLUTION

B → D → A → C

B. Base weight (inverse selection probability)

D. Non-response adjustment

A. Calibration adjustment

C. Weight trimming (applied last, if needed)

▶ Key Insight

Logic: Start with design → correct for who responded → align with benchmarks → trim extremes. Some methodologists trim before calibration—both are defensible.

Final Quiz: Question 5 of 5

SCENARIO CHALLENGE

Your organization's Bus Factor is 1. The sole methodologist plans to retire in 6 months.

List THREE specific actions you would take immediately to raise the Bus Factor to 3 before they leave.

Action 1: _____

Action 2: _____

Action 3: _____

Time: 90 seconds

Final Quiz: Answer 5

SOLUTION — Any three of these

1. **Document everything:** Have the methodologist write detailed SOPs for all weighting procedures with comments in code
2. **Shadow training:** Assign 2–3 staff to work alongside them on the next survey cycle, learning by doing
3. **Create video walkthroughs:** Record screen captures of key processes with verbal explanations
4. **Implement version control:** Move all scripts to Git so changes are tracked and explained
5. **External training:** Send staff to courses (like this workshop!) to build parallel expertise
6. **Knowledge transfer sessions:** Weekly meetings where methodologist explains "why" behind decisions

Did you get at least 3 good ideas? ✓ Yes / × No

Final Quiz: Score Yourself

How did you do?

Score	Rating
5/5	★★★ **MASTER METHODOLOGIST** — Ready to train others!
4/5	★★ **DISTINCTION** — Solid expertise, minor review needed
3/5	★ **COMPETENT** — Good foundation, practice recommended
2/5	**DEVELOPING** — Review golden rules before implementing
0–1/5	**REVISIT** — Consider reviewing workshop materials

▶ Key Insight

Remember: The real test is not this quiz—it's whether you implement what you've learned when you return to your office.

RECOGNITION

Certificate Presentation

Honoring Your Achievement

"The Republic of Zambara thanks you for your service to statistics."

Certificate of Achievement

Honorary Zambara Statistician

This certifies that

[Your Name]

has successfully completed the

SADC Regional Training Workshop
Advanced Sampling Methods for Household Surveys

and demonstrated mastery of the Golden Rules of Survey Methodo

ZNSO
SEAL

Dr. Lindiwe Moyo

Prof. Endri Raco

★ WITH DISTINCTION ★

Certificate of Achievement

Honorary Zambara Statistician

This certifies that

[Your Name]

has achieved **DISTINCTION** (4+/5) in the final assessment

of the

SADC Regional Training Workshop
Advanced Sampling Methods

MASTER LEVEL

Certificate Presentation Ceremony

[GROUP PHOTO]

Workshop Participants
with Certificates

Please gather at the front for the official photograph

YOUR FEEDBACK MATTERS

Workshop Evaluation

Help us improve future workshops

5 minutes of your time = better training for colleagues

Complete the Workshop Evaluation

Please complete the evaluation form:

1. Scan the QR code, OR
2. Visit: `forms.sadc.int/sampling2026`
3. Complete all sections (5 minutes)
4. Submit before leaving today

We want to know:

- Which modules were most valuable?
- What should we add or remove?
- How was the pacing?
- Quality of materials and exercises?
- Suggestions for future workshops?

[QR CODE]

Scan to access evaluation form

`forms.sadc.int/sampling2026`

Complete the Evaluation for Bonus Materials

▶ Key Insight

Complete the evaluation to receive:

- Full workshop slide deck (PDF)
- All R and STATA lab scripts
- GHS 2024 practice datasets
- Golden Rules reference card (printable)
- Certificate in high-resolution PDF

Materials will be emailed within 48 hours of evaluation submission.

Your feedback directly shapes next year's workshop!

Previous participant feedback led to:

Resources and Follow-Up

Stay connected after the workshop:

Workshop Materials:

- All slides and scripts on USB drive
- Online repository (link in evaluation)
- Reference textbooks list provided
- GHS 2024 documentation package

Continued Learning:

- IHSN online courses (free)
- Survey methodology MOOCs
- Stata and R user communities
- Annual SADC statistics conference

Contact Information:

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Technical Support:
SADC Statistics Unit
`stats@sadc.int`

Alumni Network:
Join the SADC Survey Methods LinkedIn group for ongoing discussion

CLOSING REMARKS

Final Reflections

From the Workshop Team

The Journey We've Shared

Zambara Case

Five days ago, you walked into this room as experienced statisticians. Today, you leave as **survey methodologists**—professionals who understand not just *how* to calculate weights, but *why* each step matters for the quality of national statistics.

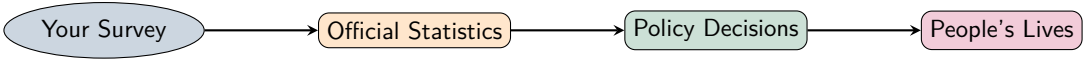
What makes this workshop different:

- We didn't just teach formulas—we taught **judgment**
- We didn't just show software—we explained **consequences**
- We didn't just present theory—we practiced with **real data**
- We didn't just cover methods—we created a **community**

Lindiwe's story is now your story.
What protocol will you build for your organization?

The Responsibility You Carry

Survey data shapes national policy:



- Budget allocations depend on poverty estimates
- Health programs target areas identified by surveys
- Education investments follow enrollment data
- Infrastructure plans use service access statistics

▶ Key Insight

When your weights are wrong, policies fail. When your methodology is sound, nations prosper. This is not just statistics—it is **public service**.

Your Call to Action

Within 30 days of returning home:

1. **AUDIT:** Complete the frame and methodology audit checklist
2. **PRESENT:** Brief your director on workshop learnings
3. **IDENTIFY:** Select a pilot survey for new methods
4. **DOCUMENT:** Begin writing SOPs for one weighting process
5. **TRAIN:** Share at least 3 golden rules with a colleague

The Lindiwe Challenge:
In 6 months, be able to present YOUR survey protocol to YOUR Minister.

Acknowledgments

This workshop was made possible by:

Partner Organizations:

- SADC Secretariat
- Statistics South Africa
- World Bank LSMS Team
- African Development Bank
- PARIS21

Data Providers:

- Stats SA (GHS 2024)
- GADM (Shapefiles)
- IHSN (Resources)

Technical Contributors:

- Prof. Endri Raço (Lead)
- National statistics offices
- Survey methodology community

Facility and Logistics:

- Conference venue staff
- IT support team
- Catering services
- All behind-the-scenes heroes

And most importantly: YOU, the participants!

The SADC Vision for Statistics

"A community of practice where every SADC member state produces high-quality, comparable household survey data that drives evidence-based policy for sustainable development."

You are now part of this vision:

- 16 SADC member states
- 400+ million people depending on good statistics
- One shared commitment to methodological excellence
- A network of trained professionals supporting each other

▶ Key Insight

When you return home, you don't return alone. You return with 50+ colleagues across SADC who speak the same methodological language and share the same commitment to quality.

"Statistics are the eyes of the nation.

With good statistics, governments see truth.
You are the optometrists of democracy."

— Workshop Director

Thank You!

Safe travels home

May your weights always sum to the population,
your standard errors be correctly estimated,
and your Bus Factor never fall below three.

SADC Regional Training Workshop
Advanced Sampling Methods for Household Surveys

March 2026 · Johannesburg, South Africa

See you next year!